

THE DISTRIBUTION OF THE LARGEST NEW EIGENVALUES OF RANDOM COVERS OF GRAPHS

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ABSTRACT. For a connected base graph B , the eigenvalues of a random degree k cover of B split into those of B and $(k - 1)|V(B)|$ new eigenvalues, and the cover is *one-sided Ramanujan* if every new eigenvalue is at most the spectral radius $\rho(B)$ of the universal cover of B . For random regular graphs, the covers of a bouquet of loops, Huang, McKenzie, and Yau recently proved that the largest new eigenvalue has Tracy-Widom fluctuations, and so about 83% of them are one-sided Ramanujan. From 1.4×10^8 samples across ten families of random covers, we conjecture the general law: the largest eigenvalue of each isotypic block follows the Tracy-Widom law of the symmetry class $\beta \in \{1, 2, 4\}$ selected by the Frobenius-Schur type of its representation, centered at $\rho(B)$ and on a scale fixed, with no free parameter, by the square root edge of the spectral measure of the universal cover. The limiting probability that a random cover is one-sided Ramanujan is then

$$\prod_{[\rho] \neq 1} F_{\beta(\rho)}(0),$$

a product over the non-trivial representations up to conjugation. With nothing fitted, the data matches the predicted scales to about one percent. We also find that the leading finite size correction is a shift of order $1/n$, which explains the 52% conjectured by Miller, Novikoff, and Sabelli, and that the far tails consist of localized eigenvalues of rare defect subgraphs, whose values and rates we compute.

1. INTRODUCTION

Let B be a finite connected graph. A degree k cover (or k -lift) H of B is obtained by assigning to each edge of B a permutation in S_k , and the eigenvalues of H split into the *old* eigenvalues, which are those of B , and the remaining $(k - 1)|V(B)|$ *new* eigenvalues (Definition 2.1). Random covers were introduced as a combinatorial construction of expanders by Amit and Linial [10, 11] (for expanders see the survey [57]), and the largest new eigenvalue λ_{new} of a random cover is governed by the spectral radius $\rho(B)$ of the universal cover of B , the infinite tree that covers B . Friedman [37] conjectured that $\lambda_{\text{new}} \leq \rho(B) + \epsilon$ asymptotically almost surely, the relative Alon conjecture, and after bounds of Linial and Puder [72], of Lubetzky, Sudakov, and Vu [74], and of Puder [93], it was proven for regular base graphs by Friedman and Kohler [42, 41] and in general by Bordenave [16] (see also [18, 19]), with a sharp quantitative form due to Chen, Garza-Vargas, Tropp, and van Handel [27, 28] (see the surveys [104, 105, 17]). No infinite family of graphs sharing the universal cover of B does better, by the Alon-Boppana bound (see Nilli [88, 89] and Friedman [36]) and its generalization by Friedman and Tillich [43], and so we call a cover *one-sided Ramanujan* if every new eigenvalue is at most $\rho(B)$. The relative Alon theorem places λ_{new} at $\rho(B)$ to first order, but says

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nothing about which side of $\rho(B)$ it falls on. The question of this paper is the finer one: how does λ_{new} fluctuate around $\rho(B)$, and what is the probability that a random cover is one-sided Ramanujan?

When B is the bouquet of $d/2$ loops on a single vertex, a random degree k cover of B is the permutation model of a random d -regular graph on k vertices, the universal cover is the d -regular tree, $\rho(B) = 2\sqrt{d-1}$, and the question is the one asked by Sarnak [97]: is a positive proportion of random d -regular graphs Ramanujan? Writing $d = \lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_V$ for the adjacency eigenvalues of a d -regular graph on V vertices, Alon [3] conjectured that $\lambda_2 \leq 2\sqrt{d-1} + \epsilon$ asymptotically almost surely, Friedman, Kahn, and Szemerédi [40, 35] proved the first $O(\sqrt{d})$ bounds, and Friedman [38] proved the conjecture. Following experiments of Novikoff [90], Miller, Novikoff, and Sabelli [84], hereafter MNS, gave numerical evidence that λ_2 fluctuates according to the $\beta = 1$ Tracy-Widom law of random matrix theory [103]. Writing μ_V and σ_V for the mean and standard deviation of λ_2 over graphs of size V , they fitted power laws

$$2\sqrt{d-1} - \mu_V \sim c_\mu V^m, \quad \sigma_V \sim c_\sigma V^s,$$

found best fit exponents m between -0.72 and -0.83 against s between -0.71 and -0.73 , and conjectured that the limiting probability of being one-sided Ramanujan is the mass of the Tracy-Widom law to the left of its mean,

$$(1.1) \quad F_1(\mu_{TW_1}) = 0.519652\dots,$$

approximately 52%. The answer is now known. After Tracy-Widom fluctuations were proven for degrees growing with V by Bauerschmidt, Huang, Knowles, and Yau [14], by He [50], and by Huang and Yau [61], and eigenvalue rigidity at fixed degree was established [60, 58], Huang, McKenzie, and Yau [59] proved the following (see also the lecture notes [62]).

Theorem 1.1. (*Huang, McKenzie, and Yau [59]*) *Let $d \geq 3$ be fixed, let G be a uniformly random simple d -regular graph on V vertices, and let $A = d(d-1)/(d-2)^2$. Then*

$$(AV)^{2/3} \left(\frac{\lambda_2(G)}{\sqrt{d-1}} - 2 \right) \rightarrow TW_1$$

in distribution as $V \rightarrow \infty$, where TW_1 is the $\beta = 1$ Tracy-Widom distribution in its original normalization. In particular, since the centering is exactly $2\sqrt{d-1}$,

$$\lim_{V \rightarrow \infty} P \left(\lambda_2(G) \leq 2\sqrt{d-1} \right) = F_1(0) = 0.8319081\dots$$

Unfolding the constants, Theorem 1.1 states that

$$(1.2) \quad \lambda_2(G) = 2\sqrt{d-1} + c(d)V^{-2/3}X_V, \quad c(d) = \left(\frac{(d-2)^2}{d(d-1)} \right)^{2/3} \sqrt{d-1},$$

where X_V converges in law to TW_1 . In particular both of MNS's exponents equal $-2/3$, and the threshold sits $|\mu_{TW_1}|/\sigma_{TW_1} = 0.951537\dots$ standard deviations to the right of the mean, where $\mu_{TW_1} = -1.2065335746\dots$ and $\sigma_{TW_1} = 1.2679830577\dots$ are the mean and standard deviation of TW_1 . The constant $c(d)$ has a transparent origin, which is the key to the constants in our conjectures. The Kesten-McKay law, the limiting spectral measure of random d -regular graphs, has density

$$(1.3) \quad \frac{d\sqrt{4(d-1)-x^2}}{2\pi(d^2-x^2)} = \left(\frac{C(d)}{\pi} + o(1) \right) \sqrt{2\sqrt{d-1}-x}, \quad C(d) = \frac{d(d-1)^{1/4}}{(d-2)^2},$$

near its right edge, and $c(d) = C(d)^{-2/3}$. In other words, $(C(d)V)^{-2/3}$ is the distance from the edge within which a square root edge of density $(C(d)/\pi)\sqrt{2\sqrt{d}-1} - x$ holds a bounded number of V eigenvalues, and at this scale λ_2 fluctuates exactly as the largest eigenvalue of the Gaussian Orthogonal Ensemble does at its own edge, where the semicircle law on $[-2, 2]$ has $C = 1$.

For covers of a base graph with more than one vertex the fluctuations are open, apart from one result on iterated 2-covers of dense random regular graphs [73]. In this paper we analyze 1.4×10^8 samples of the largest new eigenvalue across ten families of random covers, about a thousand times the data of MNS, and conjecture that the answer is governed by the representation theory of the cover and by the universal cover of the base. Our conjectures concern the following blocks of the adjacency matrix, which are described in more detail in section 2.1.

Definition 1.2. Let Γ be a finite group and let B be a finite connected graph on n vertices. A *random Γ -cover* of B assigns to each oriented edge e of B an independent uniform element $\sigma_e \in \Gamma$, with $\sigma_{\bar{e}} = \sigma_e^{-1}$ on the reversed edge, and is the graph on $V(B) \times \Gamma$ in which (u, g) is adjacent to $(v, g\sigma_e)$ for each oriented edge $e = (u, v)$. For an irreducible unitary representation ρ of Γ of dimension d_ρ , the ρ -isotypic block of the cover is the nd_ρ dimensional Hermitian matrix

$$A_\rho = \sum_{e=(u,v)} E_{uv} \otimes \rho(\sigma_e),$$

where the sum runs over the oriented edges of B and E_{uv} is the matrix unit. The *Frobenius-Schur indicator* of ρ is

$$\varepsilon(\rho) = \frac{1}{|\Gamma|} \sum_{g \in \Gamma} \chi_\rho(g^2) \in \{1, 0, -1\},$$

and ρ is called *real*, *complex*, or *quaternionic* according as $\varepsilon(\rho) = 1, 0, -1$. We write $\beta(\rho) = 1, 2, 4$ correspondingly, and $N_\rho = nd_\rho$ for real and complex ρ and $N_\rho = nd_\rho/2$ for quaternionic ρ , so that A_ρ is conjugate to an $N_\rho \times N_\rho$ Hermitian matrix over $\mathbb{R}$, $\mathbb{C}$, or the quaternions $\mathbb{H}$, respectively.

The adjacency matrix of a Γ -cover is conjugate to $\bigoplus_\rho A_\rho^{\oplus d_\rho}$, the trivial representation giving B itself, and so the new eigenvalues are those of the non-trivial blocks (Proposition 2.2). For a random degree k cover, with independent uniform permutations, the new eigenvalues are those of the single block of the standard $(k-1)$ dimensional representation of S_k , which is real. The indicator sorts the blocks into the three symmetry classes of random matrix theory, the threefold way of Dyson [34]: real symmetric (GOE, $\beta = 1$), complex Hermitian (GUE, $\beta = 2$), and quaternion self-dual (GSE, $\beta = 4$). The largest eigenvalues of these ensembles follow the Tracy-Widom laws TW_β [102, 103], and our first conjecture is that the blocks of a random cover do the same, with the scale of Theorem 1.1.

Conjecture 1.3. Fix $d \geq 3$ and a finite group Γ , and let ρ be a non-trivial irreducible representation of Γ . Let A_ρ be the ρ -isotypic block of a random Γ -cover of a uniformly random simple d -regular graph on n vertices, with $\beta = \beta(\rho)$ and N_ρ as in Definition 1.2. Then

$$\frac{\lambda_{\max}(A_\rho) - 2\sqrt{d-1}}{c(d) N_\rho^{-2/3}} \rightarrow \chi_\beta$$

in distribution as $n \rightarrow \infty$, where $c(d)$ is the constant of (1.2), and χ_β is the limit law of the rescaled largest eigenvalue of the Gaussian ensemble of class β with its semicircle law on $[-2, 2]$, namely

$$\chi_1 = TW_1, \quad \chi_2 = TW_2, \quad \chi_4 = 2^{-1/6}TW_4,$$

in the original normalization of Tracy and Widom. In particular the centering is exactly $2\sqrt{d-1}$, with no shift.

The factor $2^{-1/6}$ is not a new constant: it records the normalization Tracy and Widom chose for the GSE (Remark 2.3). The largest new eigenvalue of a Γ -cover is the maximum over its non-trivial blocks, and in our data blocks of representations that are not complex conjugates of one another behave independently, as Rivin and Sardari [95] conjectured for the bulk spectra of Cayley graphs. This yields explicit limiting Ramanujan probabilities.

Conjecture 1.4. *With the notation of Conjecture 1.3, the block maxima $\lambda_{\max}(A_\rho)$ are asymptotically independent across non-conjugate non-trivial representations, and the probability that a random Γ -cover is one-sided Ramanujan satisfies*

$$\lim_{n \rightarrow \infty} P\left(\lambda_{\max}(A_\rho) \leq 2\sqrt{d-1} \text{ for all } \rho \neq 1\right) = \prod_{[\rho] \neq 1} F_{\beta(\rho)}(0),$$

where the product runs over the non-trivial irreducible representations of Γ up to complex conjugation, and F_β is the distribution function of TW_β . Here $F_1(0) = 0.8319081\dots$, $F_2(0) = 0.96937\dots$, and $F_4(0) = 0.99857\dots$.

For example, $\mathbb{Z}_k$ has $\lfloor (k-1)/2 \rfloor$ conjugate pairs of complex characters and, for even k , the real character $(-1)^x$, and so under Conjecture 1.4 a random $\mathbb{Z}_k$ -cover is one-sided Ramanujan with limiting probability $F_2(0)^{(k-1)/2}$ for odd k and $F_1(0)F_2(0)^{(k-2)/2}$ for even k , which is $0.96937\dots$, $0.80643\dots$, and $0.93968\dots$ for $k = 3, 4, 5$.

Our third conjecture concerns covers of a fixed base graph, which need not be regular. Its scale comes from the following measure, the analog of the Kesten-McKay law.

Definition 1.5. For a finite connected graph B on n vertices, let μ_B be the average over the vertices v of B of the spectral measure of the adjacency operator of the universal cover of B at a lift of v . If its density has a square root edge at $\rho(B)$, we define the edge constant $C_B > 0$ by

$$\frac{d\mu_B}{dx} = \left(\frac{C_B}{\pi} + o(1)\right) \sqrt{\rho(B) - x} \quad \text{as } x \rightarrow \rho(B)^-.$$

The measure μ_B is the limiting spectral measure of the new eigenvalues of random covers of B . It is the Kesten-McKay law when B is regular, where $C_B = C(d)$, and it is computable for any B from a finite system of equations (section 2.2), which gives, for instance, $C_{K_5-e} = 1.7250777\dots$.

Conjecture 1.6. *Let B be a fixed connected graph on n vertices of cycle rank $e(B) - v(B) + 1$ at least two, whose measure μ_B has a square root edge at $\rho(B)$ with edge constant C_B . Let H_k be a uniformly random degree k cover of B , and let $\lambda_{\text{new}}(H_k)$ be its largest new eigenvalue. Then*

$$(C_B nk)^{2/3} (\lambda_{\text{new}}(H_k) - \rho(B)) \rightarrow TW_1$$

in distribution as $k \rightarrow \infty$. In particular

$$\lim_{k \rightarrow \infty} P(H_k \text{ is one-sided Ramanujan}) = F_1(0) = 0.8319081\dots$$

for every such base graph B .

The $\beta = 1$ law is forced by the reality of the standard representation of S_k , and for the bouquet of $d/2$ loops, where $n = 1$ and $C_B = C(d)$, Conjecture 1.6 is Theorem 1.1 for the permutation model. The hypothesis excludes only the degenerate bases: covers of a tree are disjoint copies of it and covers of a cycle are unions of longer cycles, so neither has a fluctuation theory, while vertices of degree two are harmless, since covers of a subdivision are subdivisions of covers.

1.1. Summary of the Numerical Results. Our dataset consists of ten families: random 4-regular graphs in three versions, the $\mathbb{Z}_3$, $\mathbb{Z}_4$, and $\mathbb{Z}_5$ -covers and the quaternionic block of Q_8 -covers of random 4-regular graphs, and random covers of the fixed bases K_5 , $K_5 - e$, and $K_4 - e$, with between 10^5 and 5×10^6 samples per size (Table 2.1). Table 1.1 lists, for each family at its largest size, the law and scale predicted by Conjectures 1.3, 1.4, and 1.6, and how closely the data matches them. The evidence is of three kinds.

First, each family selects the predicted law. After the best affine fit of the sample to each candidate law, which compares the shapes alone, the Kolmogorov-Smirnov distance to the predicted law is at the sampling floor and is the smallest among the candidates for every family except the slowest, $K_4 - e$ (Figure 1.1, Numerical Results 4.1 and 4.3). The tails, the mass to the left of the mean, and the skewness and kurtosis select the same laws. Second, the scales are predicted with nothing fitted. With the centering and the scale of the conjectures fixed in advance, the fitted scale over the predicted scale is within 1.2% of 1 at the largest size for every family except $K_4 - e$, where it is 1.044 and still decreasing, and the amplitudes of the fixed base families extrapolate to within 0.15% of the universal cover predictions, including the new constant $C_{K_5 - e}^{-2/3} = 0.695228 \dots$ (Numerical Results 3.1, 4.2, and 5.2). Third, the Ramanujan probabilities approach the products of Conjecture 1.4, for example 0.94011 for the $\mathbb{Z}_5$ -covers against $F_2(0)^2 = 0.93968 \dots$, and every remaining gap in the last column of Table 1.1 is accounted for, to within 3×10^{-4} in all but one family, by the finite size shift described next (Numerical Result 4.4).

The data also shows how the limits are approached, which no theorem addresses. In every family the finite size law is a shift of the whole law by a/n , with a model dependent constant a , together with a scale that converges to its predicted value. For the random 4-regular graphs conditioned to be simple the shift dominates, with $a = 4.465 \pm 0.008$, and so the Ramanujan probability at size V is $F_1(a/(c(4)V^{1/3}))$ to within 3×10^{-4} for every V from 1000 to 20,000, a function which approaches $F_1(0)$ from above only like $V^{-1/3}$ and is still 0.01 above it at $V = 8.6 \times 10^5$ (Numerical Result 3.2). The same constant reproduces the drift of the exponents that MNS fitted, and so it is the origin of their 52% (Numerical Result 3.4). The permutation model without the conditioning, the multigraph, has $a = -1.44 \pm 0.10$ and the same bulk law, and its apparent departure from Tracy-Widom, visible in every statistic built from the sample moments, comes from a thin tail of defect eigenvalues of cycle rank two, at $7/2$ and at the real root of $\lambda^3 = 8\lambda + 16$, which occur at rate $1/V$ (Numerical Result 3.5). The far tails of the fixed base families have the same origin, and here we prove that every defect subgraph occurs, and produces its eigenvalue, at the rate governed by its cycle rank (Propositions 5.4 and 5.5), conjecture that the defects account for the entire tail (Conjecture 5.6), and verify the predicted values and rates in the data (Numerical Result 5.8).

family, size	blocks	limit law	scale	γ	$\text{KS} \times 10^3$	$P(\text{Ramanujan})$
simple 4-regular, $V = 2 \cdot 10^4$	one real	TW_1	$c(4)$	1.0031	0.7 (2.2)	0.8651 (0.8319)
legacy simple, $V = 5 \cdot 10^5$	one real	TW_1	$c(4)$	1.0036	1.2 (2.3)	0.8435 (0.8319)
multigraph, $V = 5 \cdot 10^5$	one real	TW_1	$c(4)$	0.9992	1.6 (2.9)	0.8292 (0.8319)
$\mathbb{Z}_3$ -covers, $n = 5000$	one complex pair	χ_2	$c(4)$	1.0083	0.8 (2.6)	0.9679 (0.9694)
$\mathbb{Z}_4$ -covers, $n = 5000$	real, complex pair	$\max(\chi_1, \chi_2)$	$c(4)$	0.9901	1.2 (10.6)	0.8229 (0.8064)
$\mathbb{Z}_5$ -covers, $n = 5000$	two complex pairs	$\max(\chi_2, \chi'_2)$	$c(4)$	1.0055	0.7 (1.5)	0.9401 (0.9397)
Q_8 -covers, $n = 5000$	one quaternionic	χ_4	$c(4)$	1.0052	0.5 (2.7)	0.9983 (0.9986)
covers of K_5 , $V = 10^4$	one real	TW_1	0.83268	1.0091	0.7 (2.2)	0.8328 (0.8319)
covers of $K_5 - e$, $V = 10^4$	one real	TW_1	0.69523	1.0119	0.6 (2.1)	0.8292 (0.8319)
covers of $K_4 - e$, $V = 10^4$	one real	TW_1	0.21252	1.0441	2.8 (2.6)	0.7949 (0.8319)

TABLE 1.1. The main results at a glance, each family at its largest size. The columns list the non-trivial blocks (classes up to conjugation), the predicted limit law and scale constant c , where $c(4) = 0.83268$ and the fixed base constants are $C_B^{-2/3}$, the fitted scale over the predicted scale with nothing fitted (Definition 2.4), the Kolmogorov-Smirnov distance after the affine fit to the predicted law, with the smallest distance to any other candidate law in parentheses, and the fraction of samples that are one-sided Ramanujan, with the conjectured limit in parentheses. The typical distance of a sample from its own law is 0.4 , 0.9 , and 2.7×10^{-3} for 5×10^6 , 10^6 , and 10^5 samples. $K_4 - e$, the slowest family, is not yet closest to TW_1 (section 5.3).

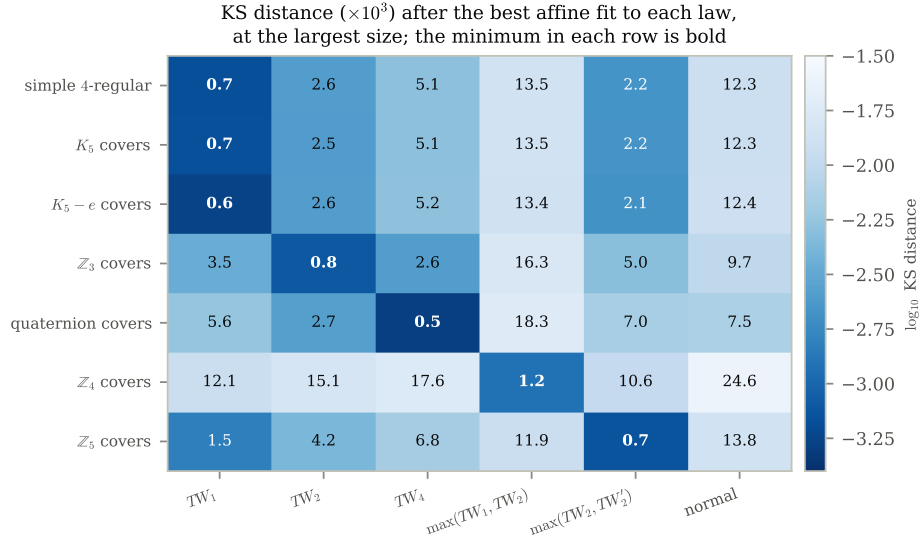


FIGURE 1.1. The selection of the limit law. Kolmogorov-Smirnov distance, times 10^3 , between each family at its largest size and each candidate law, after the best affine fit of the sample to that law, so that only the shape is compared. The minimum in each row, in bold, is the law predicted by Conjectures 1.3, 1.4, and 1.6. The typical distance of a sample from its own law is 0.4×10^{-3} at 5×10^6 samples and 0.9×10^{-3} at 10^6 .

1.2. Relation to Previous Work. The rigorous theory of the largest new eigenvalue of random covers, described above, determines its location, $\rho(B)$, and not its fluctuations, although Friedman [39] suggested from a formal expansion of the graph zeta function that the limiting probability of being Ramanujan exceeds one half, for random regular graphs and for random covers of a regular base alike, without identifying the constant. The fluctuations are proven only in two settings. For random regular graphs, the covers of a bouquet, they are Theorem 1.1, the case of Conjecture 1.6 in which B is a bouquet of loops. For iterated 2-covers of dense random regular graphs, Liu and Zou [73] proved that the largest new eigenvalue is a maximum of independent Tracy-Widom variables, the one proven instance of the maximum structure behind Conjecture 1.4. The block of the real character of $\mathbb{Z}_2$, the adjacency matrix with independent random signs, is the model of Sodin [99], whose Tracy-Widom proof was withdrawn (section 3.1), and nothing is known for complex or quaternionic blocks. For fixed irregular bases the only fluctuation related results are the Alon type theorems, of Friedman and Kohler and Bordenave for random covers and of Brito, Dumitriu, and Harris [24] for random bipartite biregular graphs.

The selection of the symmetry class by the representation type is known for bulk statistics. Gamburd, Jakobson, and Sarnak [44] observed GOE and GSE level spacings in the real and quaternionic irreducible representations of random elements of the group ring of $SU(2)$, and proved the corresponding Gaussian limit as the number of generators grows, Gamburd, Lafferty, and Rockmore [45] observed GOE and GSE spacings in the irreducible components of quantized cat maps, and Rivin and Sardari [95] split the spectra of random Cayley graphs of $SL_2(\mathbb{Z}/p\mathbb{Z})$ by irreducible representation. In quantum chaos, a time reversal invariant billiard with only a threefold symmetry shows GUE statistics [71, 68], the indicator governs the spacing statistics of desymmetrized spectra [66], and the quaternionic block of a quantum graph with the symmetry of the quaternion group shows GSE statistics without spin [67], which was then realized in a microwave experiment [94]. As k grows, the character blocks of $\mathbb{Z}_k$ -covers approach the regular graphs with independent uniform phases on the edges, whose bulk statistics Oren, Godel, and Smilansky [91, 92] compared with the GUE. For random covers of hyperbolic surfaces, the relative spectral gap is $\frac{1}{4} - \epsilon$ [76, 54, 77], with a polynomial rate [52]. For smooth linear spectral statistics of the Laplacian twisted by a character of the fundamental group, Naud [87] proved that the variance is that of the GOE for a real character and of the GUE otherwise, with extensions to higher dimensional twists [86, 78], the GSE appearing for $SU(2)$ twists [86], while for Weil-Petersson random surfaces the GOE, GUE, and GSE statistics appear for the Laplacian, for twisted Laplacians, and for Dirac operators [96, 81]. At the bottom of the spectrum, the $\beta = 1$ Tracy-Widom law has been conjectured for the first new eigenvalue of random covers and of Weil-Petersson surfaces [52, 53]. None of this work concerns the spectral edge of a graph or a representation dependent Tracy-Widom law.

The previous numerical work at the edge concerns random regular graphs, Cayley graphs, and random covers of a few fixed regular bases. Novikoff [90] and MNS [84] sampled λ_2 of random regular graphs, at 1000 graphs per size in most cases, and fitted power laws to its moments, which led to the 52% of (1.1), while Jakobson, Miller, Rivin, and Rudnick [63] found GOE statistics in the bulk. Lafferty and Rockmore [69] computed the second eigenvalues of Cayley graphs of $SL_2(\mathbb{F}_p)$ separately in different families of representations, and Rivin and Sardari [95] computed the top eigenvalue

in each representation of random Cayley graphs of $\mathrm{SL}_2(\mathbb{Z}/p\mathbb{Z})$ and reported the proportions that are Ramanujan, without fitting a law. Lubetzky, Sudakov, and Vu [74] simulated one thousand random covers per size of three cubic Ramanujan graphs, observed a stable proportion of Ramanujan covers, and conjectured that it converges to a constant in $(0, 1)$, and Crawford, Marchette, Maxwell, and Mendelson [31] studied the spectra of random graphs with a fixed equitable partition. None of these experiments identified the limit law of the largest new eigenvalue of a cover, its dependence on the representation, its scale, or the limiting Ramanujan probability.

1.3. Contributions. The contributions of this paper are the following.

First, we give evidence that the Frobenius-Schur type selects the Tracy-Widom law at the spectral edge of every isotypic block, for all three types, which extends the bulk phenomenon above to the edge, and Theorem 1.1 to the isotypic blocks of random covers of random regular graphs (Conjecture 1.3).

Second, we identify the scale of every block with no free parameter. The constant $c(d)$ of Theorem 1.1 is the square root edge constant of the Kesten-McKay law, and the same rule predicts the scale of every block over a regular base, including the quaternionic block once TW_4 is normalized like TW_1 and TW_2 , and of covers of a fixed irregular base through the universal cover, a constant which, for $K_5 - e$, the data matches to 0.15% (Conjecture 1.6).

Third, the limiting Ramanujan probability of a random cover is an explicit product of Tracy-Widom values over its representations (Conjecture 1.4), which identifies the constant conjectured by Lubetzky, Sudakov, and Vu, and which is a representation theoretic invariant of the family rather than one number, strictly between 0 and 1. Our covers of $K_5 - e$ and $K_4 - e$ give, to our knowledge, the first evidence of Tracy-Widom fluctuations around the algebraic threshold $\rho(B)$ of an irregular base.

Fourth, we measure the finite size law, a shift of order $1/n$ with a model dependent constant, which explains the 52% of MNS quantitatively and shows that Ramanujan probabilities converge only like $n^{-1/3}$, so that an experiment at accessible sizes measures F_β at a shifted point, and that the shift, not the scale, separates the fast and slow families (Numerical Results 3.2 and 3.4).

Fifth, we explain the far tails: two propositions prove the rate and the location of every localized defect, a completeness conjecture predicts that nothing else occurs, and a catalogue of defect eigenvalues matches every far outlier in our data but one, which also shows that the permutation multigraph model, whose moments behave qualitatively differently, has the same bulk law (Numerical Result 3.5).

Methodologically, we compare each sample with its conjectured law directly, with the centering and scale fixed in advance, evaluate the Tracy-Widom laws to near machine precision as Fredholm determinants, and rely on quantile statistics, which the defects cannot move.

Conjectures 1.3, 1.4, and 1.6, without the constants $c(d)N_\rho^{-2/3}$ and $(C_{Bnk})^{-2/3}$, date to 2018 and were formulated jointly with Lior Silberman, following a suggestion of Peter Sarnak. All of the data reported here was generated by 2023, before the appearance of [59], with every family but the fixed base ones predating [61] as well, and so the pipeline was in no way tuned to the theorems it now confirms. The regular graph data of section 3 thus serves as a control: the pipeline that recovers the proven law and its constants, with nothing fitted, is the pipeline that carries the cover results. Throughout, we study the one-sided question only, and make no claims about the joint behavior of the smallest new eigenvalue.

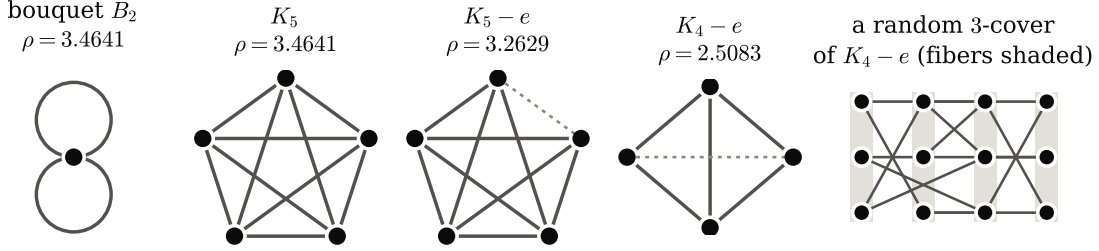


FIGURE 2.1. The base graphs of the dataset, with the spectral radius ρ of each universal cover: the bouquet of two loops, whose random covers are the permutation model of random 4-regular graphs, the complete graph K_5 , and the irregular graphs $K_5 - e$ and $K_4 - e$ (the removed edge dotted). Right: a random 3-cover of $K_4 - e$, drawn with the fiber over each base vertex as a column.

1.4. Outline. In section 2 we define the random covers and their blocks, the limit laws and the scale constants, the statistics we compute, and the dataset. Section 3 is the regular graph control experiment: the parameter-free convergence to Theorem 1.1, the finite size law and its $1/V$ shift, the origin of MNS's 52%, and the multigraph model. Section 4 presents the evidence for Conjectures 1.3 and 1.4 across abelian and quaternion covers, and section 5 the evidence for Conjecture 1.6 on the three fixed base families, together with the propositions and the conjecture on the far tails. Section 6 collects open problems.

2. THE MODEL, THE LIMIT LAWS, AND THE STATISTICS

2.1. Random Covers and Their Isotypic Blocks. We begin with the definitions used throughout.

Definition 2.1. Let B be a finite connected graph on n vertices, possibly with loops and multiple edges. A *degree k cover* of B is determined by a voltage assignment σ giving each oriented edge $e = (u, v)$ of B a permutation $\sigma_e \in S_k$, with $\sigma_{\bar{e}} = \sigma_e^{-1}$ on the reversed edge: the cover H has vertex set $V(B) \times \{1, \dots, k\}$, with (u, i) adjacent to $(v, \sigma_e(i))$ for each edge $e = (u, v)$. A *random cover* of degree k assigns independent uniform permutations to the edges of B . The eigenvalues of B are eigenvalues of H , the *old* eigenvalues, and the remaining $n(k - 1)$ eigenvalues, counted with multiplicity, are the *new* eigenvalues, with largest $\lambda_{\text{new}}(H)$. The cover is *one-sided Ramanujan* if $\lambda_{\text{new}}(H) \leq \rho(B)$, where $\rho(B)$ is the spectral radius of the adjacency operator of the universal cover of B .

A random Γ -cover (Definition 1.2) is the special case in which the fiber is Γ itself and the voltages act by right multiplication, so that $k = |\Gamma|$ and (u, g) is adjacent to $(v, g\sigma_e)$. Figure 2.1 shows the four base graphs of our dataset and a random cover. The following decomposition is standard.

Proposition 2.2. *The adjacency matrix of a Γ -cover H of B is unitarily conjugate to*

$$(2.1) \quad \bigoplus_{\rho \in \text{Irr}(\Gamma)} A_{\rho}^{\oplus d_{\rho}},$$

where A_ρ is the block of Definition 1.2, and the block of the trivial representation is the adjacency matrix A_B of B . In particular the new eigenvalues of H are the eigenvalues of the blocks A_ρ with ρ non-trivial, each with multiplicity d_ρ .

Proof. The adjacency matrix of H acts on $\mathbb{C}^{V(B)} \otimes \mathbb{C}[\Gamma]$ as $\sum_{e=(u,v)} E_{uv} \otimes R(\sigma_e)$, where R is the right regular representation, and R is unitarily equivalent to $\bigoplus_\rho \rho^{\oplus d_\rho}$. See [49, Section 2] for this decomposition, and [2] for its use on group based lifts. $\square$

The same argument applies to a random degree k cover with $\Gamma = S_k$ acting on $\{1, \dots, k\}$: the permutation representation splits as the trivial representation plus the standard $(k-1)$ dimensional representation, and the new spectrum is the single block of the standard representation, of dimension $n(k-1)$. Note that the blocks of complex conjugate representations satisfy $A_{\bar{\rho}} = \overline{A_\rho}$ and have identical spectra, and so the distinct sources of randomness are indexed by the non-trivial representations up to conjugation. The quotient operators of this decomposition also appear for quantum graphs and in graph zeta functions [13, 100].

Our families realize all three types. For $\Gamma = \mathbb{Z}_k$ the non-trivial characters $\chi_j(x) = e^{2\pi i j x / k}$ give n dimensional blocks with entries $\chi_j(\sigma_e)$ on the edges of the base, complex Hermitian in general and real symmetric exactly when $\chi_j = \pm 1$ is real valued. Thus $\mathbb{Z}_3$ has one conjugate pair of complex characters, and the largest new eigenvalue of a random $\mathbb{Z}_3$ -cover is the largest eigenvalue of a single block of class $\beta = 2$, $\mathbb{Z}_4$ has the real character $(-1)^x$ together with one conjugate pair, so that its largest new eigenvalue is the maximum of a block maximum of class $\beta = 1$ and one of class $\beta = 2$, and $\mathbb{Z}_5$ has two conjugate pairs, giving the maximum of two block maxima of class $\beta = 2$ with identical marginal distributions. The quaternion group Q_8 has four real characters and one two dimensional quaternionic representation, whose real form is four dimensional, and our quaternion family samples this real form directly: a $4n$ dimensional real matrix with each distinct eigenvalue repeated by the quaternionic structure, and so $N_\rho = n$. Together with the random covers of the bouquet of two loops (random 4-regular graphs), of K_5 , of K_5 minus an edge, and of K_4 minus an edge, these are the ten families of sections 3 through 5.

2.2. The Limit Laws and the Scale Constants. By TW_β we mean the Tracy-Widom law in the original normalization of Tracy and Widom [102, 103], with distribution function F_β , mean μ_{TW_β} , and standard deviation σ_{TW_β} , and by $\bar{T}\bar{W}_\beta$ the standardization to mean 0 and variance 1. The constants we use are

	β	μ_{TW_β}	σ_{TW_β}	$ \mu_{TW_\beta} /\sigma_{TW_\beta}$	$F_\beta(\mu_{TW_\beta})$
(2.2)	1	-1.2065335746	1.2679830577	0.9515376	0.519652
	2	-1.7710868074	0.9017731382	1.9640048	0.515016
	4	-2.3068848932	0.7195302084	3.2060987	0.511069

following [21, 20], and the values $F_1(0) = 0.8319081\dots$, $F_2(0) = 0.96937\dots$, $F_4(0) = 0.99857\dots$ of the distribution functions at the origin. We evaluate F_β as Fredholm determinants by Gauss-Legendre quadrature, following Bornemann [21], which is accurate to near machine precision and so usable deep into both tails, and which reproduces every constant in (2.2) to the digits shown. Figure 2.2 shows the three densities, and the maximum laws of section 4.3.

The scales in Conjecture 1.3 refer to the Gaussian ensembles normalized alike: $N^{2/3}(\lambda_{\max} - 2) \rightarrow \chi_\beta$ for the Gaussian ensemble of class β and dimension N over $\mathbb{R}$, $\mathbb{C}$,

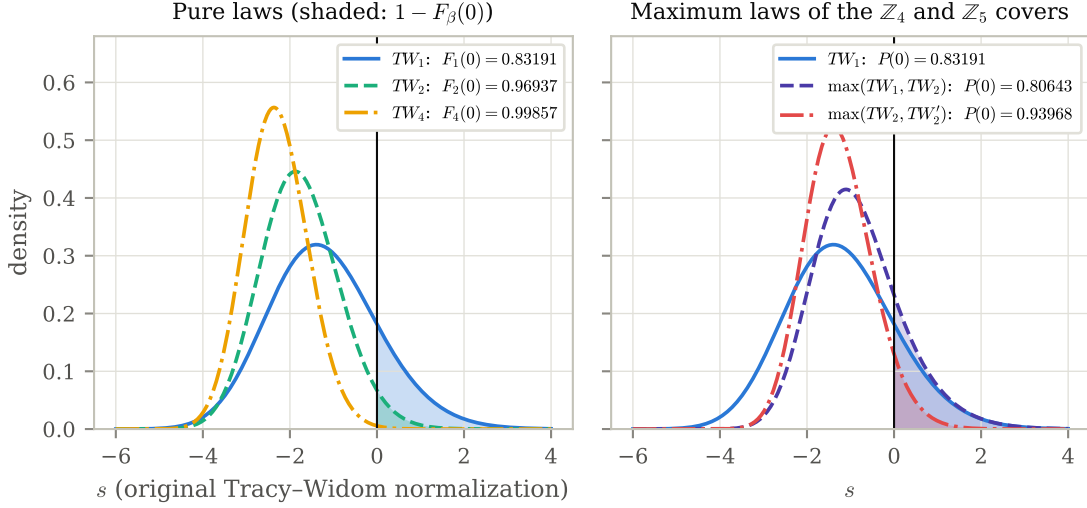


FIGURE 2.2. Left: the densities of TW_1 , TW_2 , and TW_4 in the original normalization, with the mass $1 - F_\beta(0)$ to the right of the origin shaded, the limiting probability that a block of class β is not Ramanujan. Right: the densities of the maximum laws of the $\mathbb{Z}_4$ and $\mathbb{Z}_5$ -covers, against TW_1 .

or $\mathbb{H}$, normalized so that its semicircle law is supported on $[-2, 2]$. For $\beta = 1, 2$ this is the definition of TW_β , and for $\beta = 4$ it is the following.

Remark 2.3. Tracy and Widom normalized the GSE differently [103, Section VII]. In the notation of Bornemann's review, the soft edge limit of the GSE of complex dimension $2N$ is taken at $\sqrt{4N}$ on the scale $2^{-1/2}(2N)^{-1/6}$ [20, (2.4)], and $F_4(s)$ is its distribution function evaluated at $\sqrt{2}s$ [20, (2.10)]. Rescaling the edge $\sqrt{4N}$ to 2 gives

$$\chi_4 = 2^{-2/3}\sqrt{2}TW_4 = 2^{-1/6}TW_4.$$

A direct simulation of 2×10^4 samples of this ensemble with $N = 200$ gives a standard deviation of 0.6417 for $N^{2/3}(\lambda_{\max} - 2)$, against $2^{-1/6}\sigma_{TW_4} = 0.6410$, while the same simulation of the GOE and GUE gives 1.2620 and 0.9054, against $\sigma_{TW_1} = 1.2680$ and $\sigma_{TW_2} = 0.9018$.

The scale constants themselves come from the square root edge of the limiting spectral measure, as in (1.3). For a base graph B , the measure μ_B of Definition 1.5 is computable from the branch values $y_{u \rightarrow v}(z)$, the solution with positive imaginary part for $\text{Im } z > 0$ of

$$(2.3) \quad y_{u \rightarrow v}(z) = \left(z - \sum_{w \sim v, w \neq u} y_{v \rightarrow w}(z) \right)^{-1}$$

over the directed edges of B , since the Green's function of the universal cover at a lift of v is $G_v(z) = (z - \sum_{w \sim v} y_{v \rightarrow w}(z))^{-1}$ and $d\mu_B/dx = -\frac{1}{\pi n} \sum_v \text{Im } G_v(x + i0)$. Solving (2.3) by Newton's method, continued from the upper half plane to within 10^{-12} of the

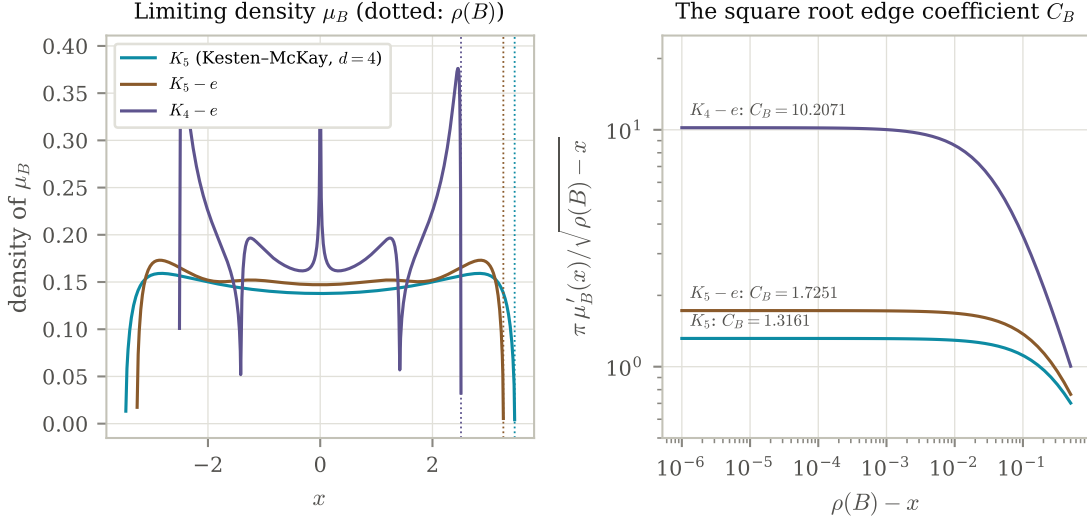


FIGURE 2.3. Left: the limiting densities μ_B of the new eigenvalues of random covers of K_5 (the Kesten-McKay law with $d = 4$), $K_5 - e$, and $K_4 - e$, computed from the branch recursion (2.3), with the edges $\rho(B)$ dotted. Right: $\pi \mu'_B(x) / \sqrt{\rho(B) - x}$ against the distance to the edge, which converges to the constant C_B of (2.4). The square root regime of $K_4 - e$ is an order of magnitude narrower than those of K_5 and $K_5 - e$.

real axis, and extrapolating $\pi (d\mu_B/dx) / \sqrt{\rho(B) - x}$ to the edge (Figure 2.3), we obtain

	B	$\rho(B)$	C_B	$C_B^{-2/3}$
(2.4)	K_5	$2\sqrt{3} = 3.4641016$	$3^{1/4} = 1.3160740$	$3^{-1/6} = 0.8326832$
	$K_5 - e$	3.2628765	1.7250777	0.6952285
	$K_4 - e$	$\sqrt{1 + 2\sqrt{7}} = 2.5082868$	10.2071395	0.2125188

where the K_5 row recovers $C(4)$ and $c(4)$ of (1.2) and (1.3). By Conjecture 1.6, the largest new eigenvalue of a random cover of B on $V = nk$ vertices is $\rho(B) + C_B^{-2/3} V^{-2/3} X$ with X asymptotically TW_1 . Note that the square root law for $K_4 - e$ sets in only within about 10^{-3} of its edge, against 10^{-2} for $K_5 - e$, which will matter in section 5.3.

2.3. The Statistics. For each family and size we record the sample mean μ_V , standard deviation σ_V , skewness, and excess kurtosis of the largest new eigenvalue λ , the Kolmogorov-Smirnov distance between the standardized sample and each of $\widetilde{TW}_1, \widetilde{TW}_2, \widetilde{TW}_4$ and the standard normal, the fraction of samples below the Ramanujan threshold, and the fraction below the sample mean. The last statistic is MNS's discriminator: the standardized Tracy-Widom laws are nearly indistinguishable by their overall shape, however their masses to the left of the mean, listed in (2.2), differ in the third decimal, and with 10^6 to 5×10^6 samples the binomial standard error is between 2×10^{-4} and 5×10^{-4} , separating the four candidates by tens of standard errors. At MNS's 1000 samples per size the same separation is a third of a standard error (Figure 2.4).

The relevance of the ratio $g_V = (\rho - \mu_V) / \sigma_V$ of the mean gap to the standard deviation, with ρ the family's threshold, is the following elementary observation of

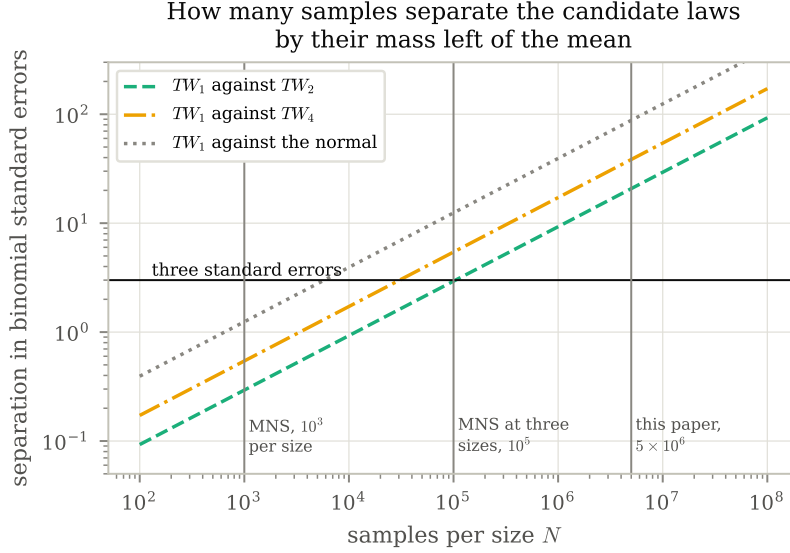


FIGURE 2.4. The separation, in binomial standard errors, of the mass of TW_1 to the left of its mean from those of TW_2 , TW_4 , and the normal law, against the number N of samples per size. With $N = 1000$, as in most of MNS's sizes, TW_1 and TW_2 are separated by 0.3 standard errors, and with our 5×10^6 by 21.

MNS. For a d -regular family, the probability of being one-sided Ramanujan is

$$P\left(\lambda \leq 2\sqrt{d-1}\right) = P\left(\frac{\lambda - \mu_V}{\sigma_V} \leq g_V\right),$$

and so if the standardized law converges to $\widetilde{TW}_1$, then

$$(2.5) \quad \lim_{V \rightarrow \infty} P\left(\lambda \leq 2\sqrt{d-1}\right) = \begin{cases} F_1(\mu_{TW_1}) = 0.519652\dots & \text{if } g_V \rightarrow 0 \\ \widetilde{F}_1(r) & \text{if } g_V \rightarrow r \in (0, \infty) \\ 1 & \text{if } g_V \rightarrow \infty \end{cases}$$

MNS's power law fits pointed to the first case. The Tracy-Widom scaling (1.2), with no shift in the centering, forces the middle case with $r = |\mu_{TW_1}|/\sigma_{TW_1}$, where $\widetilde{F}_1(r) = F_1(0)$, and this is what our data shows and what Theorem 1.1 now proves.

Standardizing by the sample mean and standard deviation discards the two numbers the conjectures predict, and so we also compare each sample with its conjectured law directly.

Definition 2.4. Consider a family whose conjectured law is $\lambda \approx \rho + cN^{-2/3}\chi$, where χ is χ_β , a maximum law, or TW_1 , and N is the block dimension ($N = V$ for the regular and fixed base families, and $N = n$ for the character and quaternion blocks over a base on n vertices), with $c = c(4)$ for every 4-regular family and $c = C_B^{-2/3}$ for a fixed base B . The *parameter-free variable* of a sample λ is

$$s = \frac{(\lambda - \rho)N^{2/3}}{c}.$$

Let s_p and χ_p be the p -quantiles of the sample of s and of the law of χ , and let $\alpha_N + \gamma_N \chi_p$ be the least squares line through the points (χ_p, s_p) for $p = 0.05, 0.06, \dots, 0.95$. The *scale ratio* is γ_N , and the *location coefficient* is $a_N = -\alpha_N c N^{1/3}$.

family	sizes	samples	model
simple 4-regular	$100 \leq V \leq 20,000$ (8)	5,000,000	two permutations, repaired
simple 4-regular (legacy)	$5 \times 10^4 \leq V \leq 5 \times 10^5$ (4)	100,000	two permutations, rejection
permutation multigraph	$5 \times 10^4 \leq V \leq 5 \times 10^5$ (4)	100,000	loops and multiple edges kept
$\mathbb{Z}_k$ -covers, $k = 3, 4, 5$	$100 \leq n \leq 5000$ (6 each)	1,000,000	random simple 4-regular base
Q_8 -covers, quaternionic block	$100 \leq n \leq 5000$ (6)	1,000,000	random simple 4-regular base
covers of K_5	$100 \leq V \leq 10,000$ (7)	5,000,000	fixed base K_5
covers of $K_5 - e$	$100 \leq V \leq 10,000$ (7)	5,000,000	fixed irregular base
covers of $K_4 - e$	$100 \leq V \leq 10,000$ (7)	1,000,000	fixed base of cycle rank two

TABLE 2.1. The dataset: for each family, the range of sizes (with the number of sizes), the number of samples per size, and the model. Here V is the number of vertices, and n the number of vertices of the random base graph.

The scale ratio should tend to 1, and the location coefficient is defined so that the fitted law of λ is

$$(2.6) \quad \lambda \approx \rho - \frac{a_N}{N} + \gamma_N c N^{-2/3} \chi.$$

Standard errors of γ_N and a_N come from ten batches in generation order. The Kolmogorov-Smirnov distance of $(s - \alpha_N)/\gamma_N$ to the conjectured law, the *distance after the affine fit*, then measures the shape alone, and it is unaffected by the far tails of section 5, which move the sample moments but not the quantiles.

2.4. The Dataset. The dataset consists of 1.4×10^8 samples across the ten families of Table 2.1, each sample the largest new eigenvalue of one independently generated graph or cover, all with base degree $d = 4$.

The simple 4-regular series draws two independent uniform permutations π_1, π_2 of $\{1, \dots, V\}$, forms the graph with adjacency matrix $P_1 + P_1^T + P_2 + P_2^T$, and repairs it locally, drawing each permutation in turn and repeatedly permuting its entries at the positions that create a loop or a multiple edge, together with one random further position, until none remains, and it discards the rare disconnected samples. The legacy simple series instead imposes simplicity by rejection, redrawing the whole graph. Both produce simple 4-regular graphs that differ from the permutation model (MNS's family $G_{V,4}$, analyzed first by Friedman [35], and a random degree V cover of the bouquet of two loops) in a bounded number of positions, and the permutation model conditioned on simplicity is contiguous with the uniform simple 4-regular model [48, 106], to which Theorem 1.1 applies. Contiguity alone does not transfer a limit law, and it is the data of section 3 that shows both series following the law of Theorem 1.1 with its constant. The permutation multigraph series keeps the loops and multiple edges, with a loop contributing 2 to the diagonal. The abelian series draw a fresh random simple 4-regular base graph on n vertices for each batch, assign each base edge an independent uniform voltage in $\mathbb{Z}_k$, and record the largest new eigenvalue of the explicit kn -vertex cover, ten covers per base graph. A one way analysis of variance across the ten covers sharing a base gives intraclass correlations below 3×10^{-4} in absolute value at the largest size, and below 2.4×10^{-3} at every size, and so the residual randomness of the base is invisible at the fluctuation scale and the samples may be treated as independent. The quaternion series assigns each edge of a random simple 4-regular base an independent uniform element of Q_8 and records the top eigenvalue of the resulting $4n$ dimensional real block. The fixed base series assign each edge of K_5 , of K_5 minus an edge, and of

K_4 minus an edge an independent uniform permutation in S_k , and record the largest new eigenvalue of the cover.

We compute eigenvalues with the implicitly restarted Lanczos method (ARPACK), applied to the adjacency matrix shifted by a small multiple of the identity so that the algebraically largest eigenvalues dominate in magnitude. We remove old eigenvalues by comparing against the base spectrum, computed exactly, with a tolerance of 10^{-10} . The number of computed eigenvalues per graph starts at 2 and grows automatically whenever every computed eigenvalue is old. For the regular families this removes the trivial eigenvalue d and yields λ_2 . For a disconnected cover, whose spectrum contains that of B twice, removal by value discards the second copy as well, an event of probability $O(k^{1-r(B)})$ by Proposition 5.4 below, where $r(B)$ is the cycle rank of B . Two precision caveats apply. The legacy series were generated years earlier in MATLAB and are recorded on a 10^{-12} grid, coarser than machine precision. A per process seeding scheme caused seed collisions between worker tasks in the abelian series: of its 1.8×10^6 base graph batches, 452,508 (25%) are redundant copies of another batch, concentrated at the smallest sizes (55 to 81% of batches at $n \in \{100, 200\}$, and below 0.1% at $n \geq 1000$), so that the effective number of independent samples at the smallest sizes is smaller than the nominal 10^6 . Duplicated batches agree to better than 3×10^{-13} , and dropping them moves every statistic quoted in this paper by at most two units in its final displayed digit, consistent with the sampling fluctuation of the removed samples. We verified every summary statistic by independent recomputation from the raw samples. The complete dataset, the generation code, and the analysis pipeline that produces every figure and table of this paper are available at <https://github.com/enaslund/random-graphs>.

3. THE REGULAR GRAPH CONTROL EXPERIMENT

In this section we test the pipeline against Theorem 1.1 along the two simple 4-regular series, measure the finite size law, diagnose the artifact behind MNS's 52%, and show that the permutation multigraph, whose moments misbehave, has the same law in its bulk.

3.1. Convergence to the Proven Law. For $d = 4$ we have that $c(4) = 3^{-1/6} = 0.8326831\dots$, and so the proven amplitudes of the mean gap and standard deviation are

$$(3.1) \quad 2\sqrt{3} - \mu_V \approx 1.0046602 V^{-2/3}, \quad \sigma_V \approx 1.0558282 V^{-2/3}.$$

The most direct test uses no fitted parameter at all.

Numerical Result 3.1. For the simple random 4-regular graphs, with the parameter-free variable $s = (\lambda_2 - 2\sqrt{3})V^{2/3}/c(4)$ compared with TW_1 , the scale ratio γ_V decreases monotonically from 1.117 at $V = 100$ to 1.0031 ± 0.0003 at $V = 20,000$, and equals 1.0036 ± 0.0033 at $V = 500,000$ in the legacy series. A fit $c + bV^{-2/3}$ to the fitted amplitudes $\gamma_V c(4)$ extrapolates to 0.8318 to 0.8321, against $c(4) = 0.8326832$. The standardized Kolmogorov-Smirnov distance to $\widetilde{TW}_1$ falls to 8.7×10^{-4} at $V = 20,000$, against 3.9×10^{-3} , 7.8×10^{-3} , and 1.9×10^{-2} for $\widetilde{TW}_2$, $\widetilde{TW}_4$, and the normal, and the mass to the left of the mean is 0.51885 at $V = 20,000$ and 0.52011 at $V = 500,000$, against the TW_1 value 0.519652.

Figure 3.1 compares the distribution function of s with F_1 at five sizes: the scale is right, and what remains is a shift, which we analyze in subsection 3.2, after which

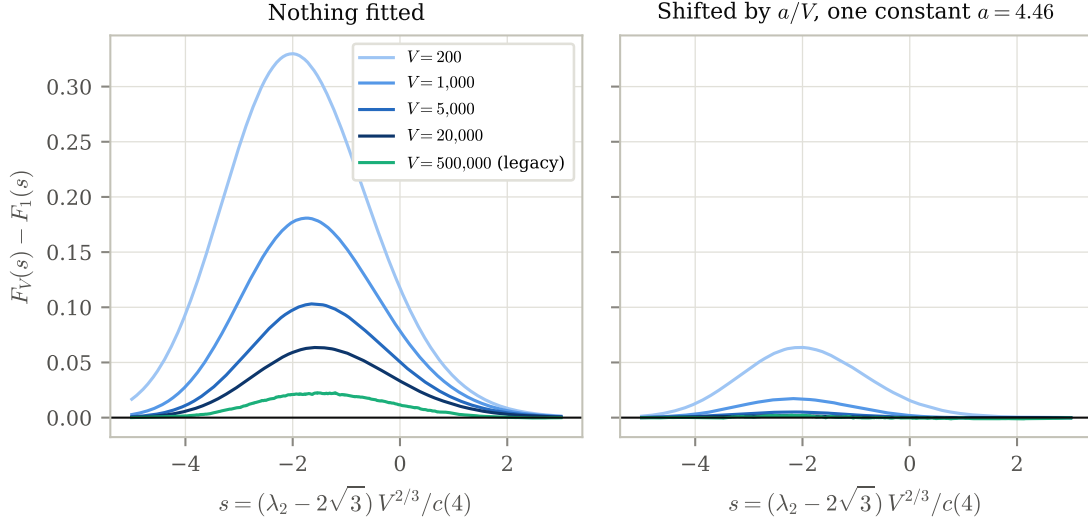


FIGURE 3.1. The random simple 4-regular graphs against Theorem 1.1 with nothing fitted: $F_V(s) - F_1(s)$, where F_V is the empirical distribution function of $s = (\lambda_2 - 2\sqrt{3})V^{2/3}/c(4)$. Left: the raw differences, a shift which shrinks with V . Right: the same after shifting every size by a/V , with the single constant a of Numerical Result 3.2.

the differences left by the scale and shape corrections are below 0.006 from $V = 5000$ on. The standardized statistics tell the same story. Figure 3.2 shows the standardized empirical density of 5×10^6 samples at $V = 20,000$ against the four standardized candidate laws, and Figure 3.3 the quantile-quantile plot of the largest legacy dataset, $V = 500,000$, against $\widehat{TW}_1$ across the probability range 10^{-4} to 0.999. The Kolmogorov-Smirnov distances of Figure 3.4 show that only the distance to $\widehat{TW}_1$ keeps falling, reaching a small multiple of the sampling floor, which is roughly 4×10^{-4} at 5×10^6 samples, and at $V = 500,000$ the Kolmogorov-Smirnov test accepts $\widehat{TW}_1$ with $p = 0.99$. In the mass to the left of the mean the competing values 0.515016, 0.511069, and 0.5 are excluded by more than fifteen binomial standard errors (Figure 4.3 in section 4), and the sample skewness and excess kurtosis at the largest sizes, 0.286 and 0.156 at $V = 20,000$, converge to the TW_1 values 0.293 and 0.165 (Figure 4.4).

Figure 3.5 shows the mean gap and standard deviation against V on logarithmic axes with the $V^{-2/3}$ guide, and Figure 3.6 shows the compensated quantities $(2\sqrt{3} - \mu_V)V^{2/3}$ and $\sigma_V V^{2/3}$, which approach the two constants of (3.1): at $V = 500,000$ the measured standard deviation is 1.685×10^{-4} against the predicted 1.676×10^{-4} , and the measured gap is 1.684×10^{-4} against the predicted 1.595×10^{-4} , the mean converging visibly more slowly than the standard deviation, by exactly the shift of Numerical Result 3.2. The constants (3.1) appear in Sodin's work [99] on a closely related model in which the matrix entries carry independent random signs. We note that the arXiv posting of [99] was withdrawn over an error in one of its lemmas, and the published version carries a 2017 erratum, so that Theorem 1.1 is the rigorous source for (1.2).

3.2. The Finite Size Law. The theorems give no rate of convergence, and the data is precise enough to measure the finite size law itself. Its leading term is the shift visible in Figure 3.1.

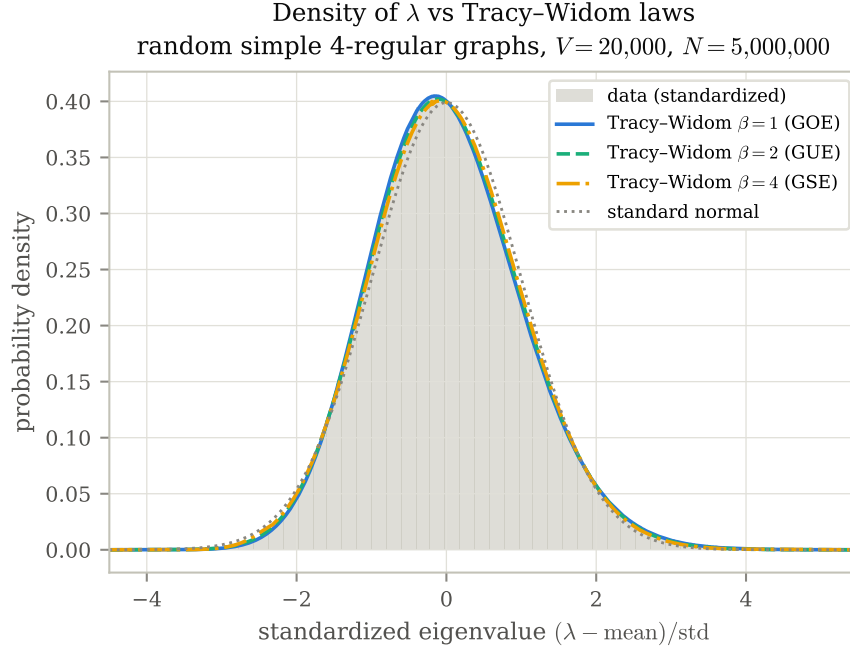


FIGURE 3.2. Standardized empirical density of the largest non-trivial eigenvalue over 5×10^6 random simple 4-regular graphs on $V = 20,000$ vertices, against the standardized Tracy-Widom $\beta = 1, 2, 4$ densities and the standard normal.

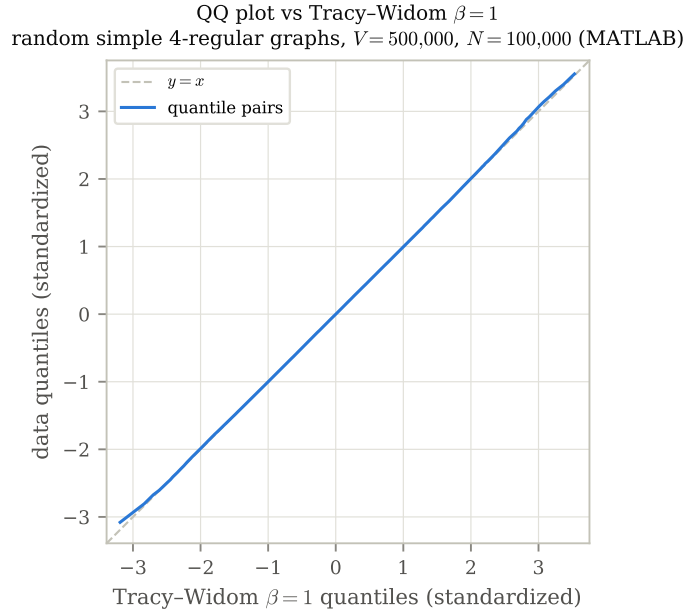


FIGURE 3.3. Quantile-quantile plot of 10^5 samples at $V = 500,000$ against $\widetilde{TW}_1$, probability range 10^{-4} to 0.999.

Numerical Result 3.2. For the simple random 4-regular graphs, the location coefficient of Definition 2.4 is $a_V = 5.24$ at $V = 100$, 4.54 at $V = 1000$, and 4.479, 4.461, and 4.453 at $V = 5000$, 10,000, and 20,000, with standard errors between 0.01 and 0.02,

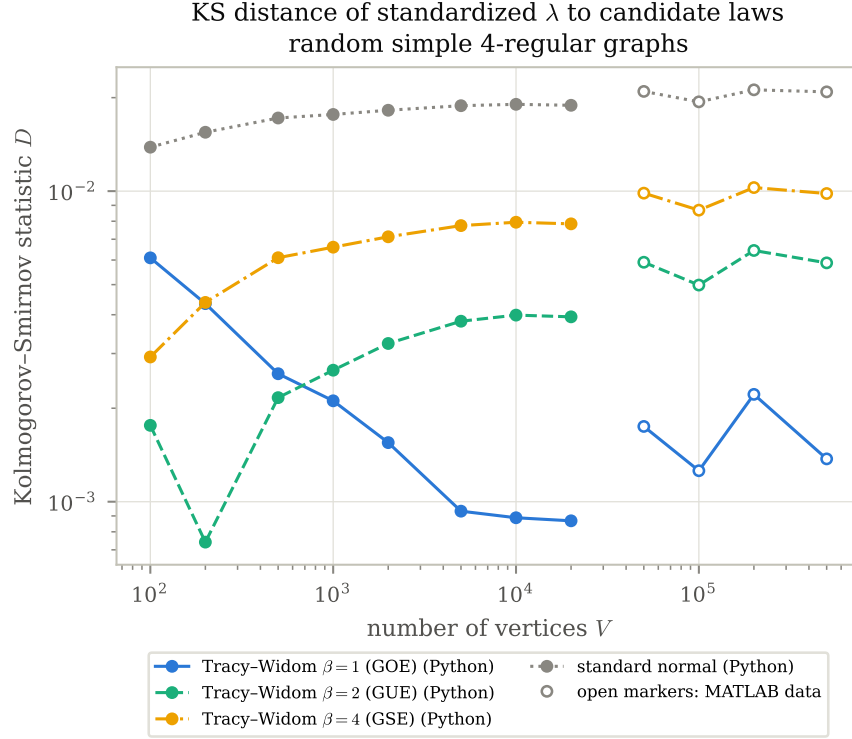


FIGURE 3.4. Kolmogorov-Smirnov distance of the standardized samples to the standardized Tracy-Widom $\beta = 1, 2, 4$ laws and the standard normal, for random simple 4-regular graphs (filled markers: 5×10^6 samples per size, open markers: legacy data with 10^5 samples). Only the distance to $\beta = 1$ keeps falling.

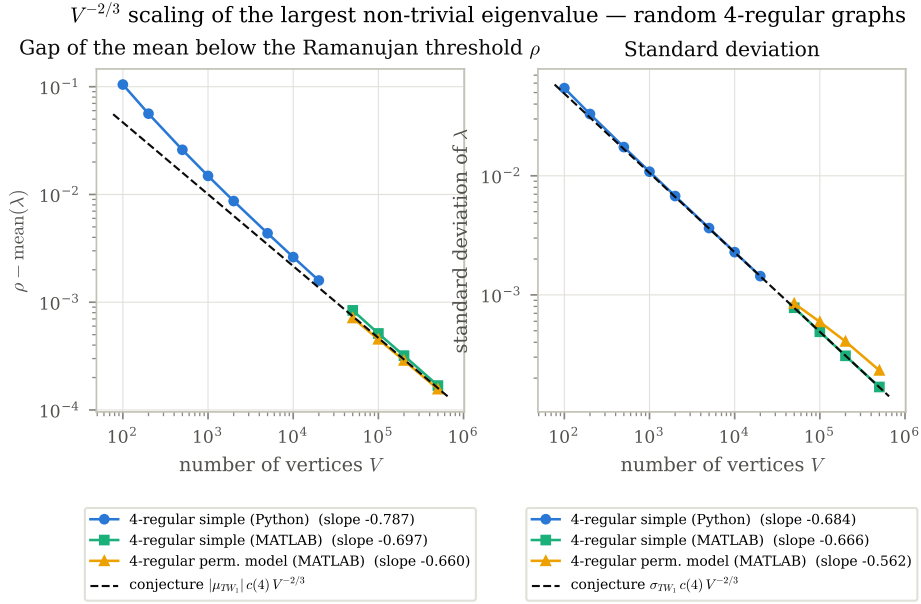


FIGURE 3.5. Mean gap $2\sqrt{3} - \mu_V$ and standard deviation against V for the three 4-regular series, with fitted slopes and the $V^{-2/3}$ Tracy-Widom line.

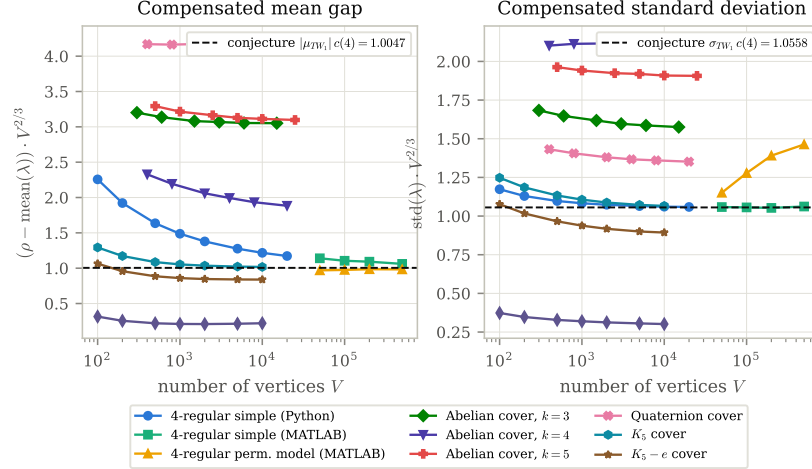
Compensated $V^{2/3}$ amplitudes — dashed: constants conjectured for random 4-regular graphs

FIGURE 3.6. Compensated scaling: $(\rho - \mu_V) V^{2/3}$ and $\sigma_V V^{2/3}$ against V for all families, where ρ is the family's Ramanujan threshold. The dashed lines are the regular graph amplitudes (3.1), which the simple 4-regular and K_5 cover series approach, while each remaining family plateaus at its own amplitude, predicted by Conjectures 1.3 and 1.6 (see sections 4 and 5).

and so, with a the weighted mean over $V \geq 5000$,

$$(3.2) \quad \lambda_2 \approx 2\sqrt{3} - \frac{a}{V} + c(4)V^{-2/3}TW_1, \quad a = 4.465 \pm 0.008.$$

With this single constant,

$$(3.3) \quad P(\lambda_2 \leq 2\sqrt{3}) \approx F_1\left(\frac{a}{c(4)V^{1/3}}\right) = F_1(0) + \frac{0.9728}{V^{1/3}} + O(V^{-2/3})$$

matches the data to within 3×10^{-4} , or 2.4 binomial standard errors, at every size from $V = 1000$ to 20,000, and to within 2.6 standard errors at the four legacy sizes. The permutation multigraph has $a = -1.44 \pm 0.10$ over its four sizes, the covers of K_5 have a between 0.01 and 0.07 from $V = 1000$ on, and the legacy simple series gives $a = 4.87 \pm 0.12$ over its four sizes.

In units of the fluctuation scale the shift is $a/(c(4)V^{1/3})$, which is 0.20 at $V = 20,000$ and 0.07 at $V = 500,000$, while the scale correction is much smaller, with $\gamma_V - 1$ decaying roughly like $V^{-2/3}$. In (3.3), $0.9728 = f_1(0)a/c(4)$, where $f_1(0) = 0.18142$ is the density of TW_1 at the origin, and so the Ramanujan probability of the simple model is still 0.01 above its limit at $V = 8.6 \times 10^5$, and reaches within 10^{-3} only at $V = 9 \times 10^8$ (Figure 3.8). For the multigraph the approach is from below, since $a < 0$, and it is within 10^{-3} at $V = 3 \times 10^7$, while the covers of K_5 , with a nearly zero, are within 10^{-3} from $V = 1000$ on. Any experiment on the Ramanujan probability at accessible sizes measures F_β at a shifted point, and the shift must be accounted for.

The shift is not a property of the degree alone. Conditioning on simplicity moves the whole law down by about $6/V$ relative to the multigraph, and the difference of 3.4 standard errors between the two simple series, which impose simplicity by local repair and by rejection, we do not regard as established. A shift of order $1/V$ is what one expects from the short cycles: in the physics literature, Metz, Parisi, and Leuzzi [83]

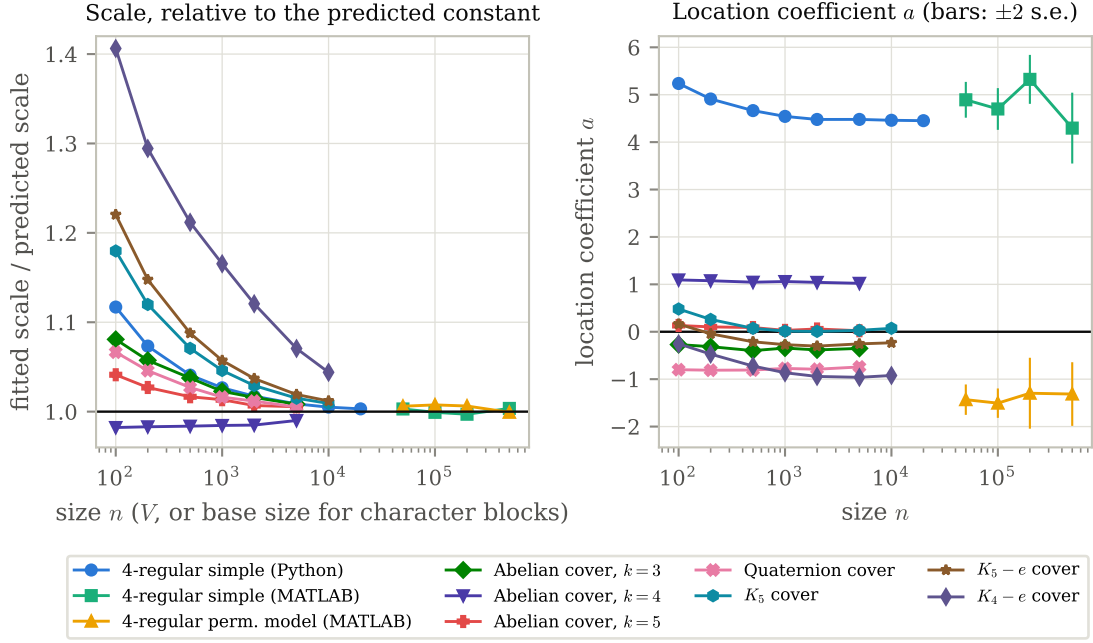


FIGURE 3.7. The two finite size parameters of Definition 2.4 for every family. Left: the scale ratio, the fitted scale over the predicted one, which is $c(4)$ for the 4-regular families, $c(4)2^{-1/6}$ in terms of TW_4 for the quaternion covers, and $C_B^{-2/3}$ of (2.4) for $K_5 - e$ and $K_4 - e$. Right: the location coefficient a , in the law $\rho - a/N + \gamma c N^{-2/3} \chi$, which settles at a family dependent constant.

computed the $1/V$ correction to the Kesten-McKay density of random regular graphs as a sum over cycle lengths, with an inverse square root singularity at the edge, which is the form of a deterministic shift of the edge at order $1/V$, and conditioning on simplicity removes the cycles of length one and two. For sparse Erdős-Rényi graphs the corresponding deterministic shift of the edge is proven [70, 25].

The third correction is to the shape. Rates at the Tracy-Widom edge have a history in random matrix theory: Choup [30] computed the Edgeworth expansion of the largest eigenvalue law of the finite GOE, Johnstone and Ma [65] proved that the rate of approach depends on the centering convention, Bornemann [22] expanded the laws of the finite Gaussian ensembles at the soft edge in terms of the derivatives of F_β , and rates of order $N^{-1/3}$ are proven for Wigner matrices [98] and sparse Erdős-Rényi graphs [25]. Write F_V for the distribution function of the standardized largest non-trivial eigenvalue and $\tilde{F}_1, \tilde{f}_1$ for the standardized Tracy-Widom distribution and density.

Numerical Result 3.3. Across the simple 4-regular family, the deviation $F_V - \tilde{F}_1$ of the standardized law is, to within 6 to 14 percent of its squared norm, proportional to the single fixed profile $(t^2 - 1) \tilde{f}_1(t)$, so that

$$F_V(t) \approx \tilde{F}_1(t) - \hat{c}(V) (t^2 - 1) \tilde{f}_1(t),$$

where the fitted amplitude $\hat{c}(V)$ decays like a power law with exponent -0.46 ± 0.04 , and the K_5 cover family lies on the same curve.

The profile is a correction of Edgeworth type signaling a skewness deficit (Figure 3.9), and it accounts for the residuals of the individual standardized statistics: the mass left

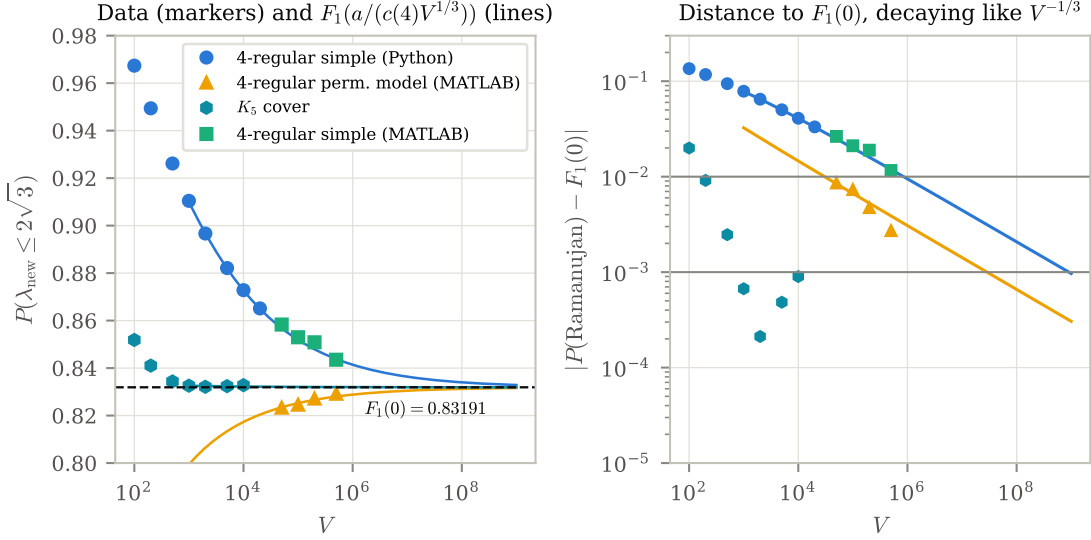


FIGURE 3.8. Left: the empirical Ramanujan probability against V for the simple, legacy simple, multigraph, and K_5 cover series (markers), with the one parameter model $F_1(a/(c(4)V^{1/3}))$ of (3.3) (lines, drawn from $V = 1000$). Right: the distance to $F_1(0)$ on logarithmic axes, extrapolated to $V = 10^9$, with the levels 10^{-2} and 10^{-3} .

The deviation of the standardized ECDF from TW_1 is a single skewness profile with power-law amplitude

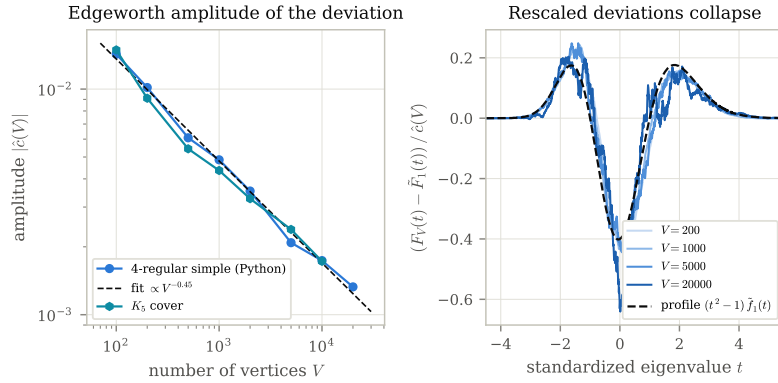


FIGURE 3.9. The shape correction. Left: the amplitude $\hat{c}(V)$ of the deviation of the standardized empirical distribution from $\widetilde{TW}_1$, for the simple 4-regular and K_5 cover families, with the fitted power law. Right: the deviations at four sizes, rescaled by $\hat{c}(V)$, collapse onto the fixed profile $(t^2 - 1)\tilde{f}_1(t)$ of a skewness deficit.

of the mean at $V = 20,000$ still sits -8.0×10^{-4} from the TW_1 value, 3.6 binomial standard errors, against an Edgeworth prediction of $-\hat{c}\tilde{f}_1(0) = -5.3 \times 10^{-4}$, and after subtracting the fitted profile the left tails of the largest datasets agree with $\widetilde{TW}_1$ to within three standard errors down to probability 6×10^{-7} .

3.3. The Origin of the 52% Conjecture. MNS reached the first case of (2.5) by fitting single power laws to the mean gap and standard deviation over their full range of sizes. Our data reproduces their fit, and (3.2) explains it. Under (3.2) the mean gap

is $AV^{-2/3} + a/V$ with $A = 1.0046602$ from (3.1), whose local exponent is

$$(3.4) \quad \frac{d \log (2\sqrt{3} - \mu_V)}{d \log V} = -\frac{2/3 + x}{1 + x}, \quad x = \frac{a}{AV^{1/3}},$$

and so approaches $-2/3$ from below only as $V^{-1/3}$.

Numerical Result 3.4. Fitting one line to $\log (2\sqrt{3} - \mu_V)$ against $\log V$ over our eight sizes $V \leq 20,000$, the range of MNS, gives a slope of -0.787 , and the same fit to $\log \sigma_V$ gives -0.684 , closely matching MNS's $d = 3$ values $m \approx -0.79$ and $s \approx -0.72$. The slopes between consecutive sizes of the mean gap drift monotonically from -0.90 at $V \approx 100$ to -0.72 at $V = 20,000$, and reach -0.70 over the last pair of legacy sizes at $V = 500,000$, along the curve (3.4), which is -0.72 at $V = 14,000$, while those of the standard deviation are within 0.01 of $-2/3$ from $V \approx 2000$ on. Fitted to the model (3.2) at our eight sizes, a single power law gives -0.764 .

The mean exponent thus appears steeper than the standard deviation exponent, exactly the pattern from which MNS concluded $g_V \rightarrow 0$ and hence 52%, and the one constant model accounts for most of the observed -0.787 , the remainder coming from the larger values of a_V at the smallest sizes (Figure 3.10). MNS saw the warning sign, observing that their best fit exponents increased monotonically as sizes were added, and were appropriately cautious. At $V \leq 20,000$ the data could do no better: by (3.3) the Ramanujan probability itself is 0.865 at $V = 20,000$, and no single size in that range is close to the limit. The ratio g_V resolves the question directly: as Figure 4.11 shows, g_V does not tend to 0 for any family, and for the simple 4-regular series it descends to 0.9993 at $V = 500,000$, approaching the Tracy-Widom ratio $|\mu_{TW_1}|/\sigma_{TW_1} = 0.951537\dots$ from above, in agreement with the middle case of (2.5) and with $F_1(0) = 0.8319081\dots$ as the limiting Ramanujan probability.

3.4. The Multigraph Model. MNS pooled models with and without loops and multiple edges, observing similar behavior at their sample sizes. At ours, the sample moments of the permutation multigraph series, identical to the legacy simple series except that loops and multiple edges are kept, differ qualitatively: at $V = 500,000$ its skewness is 74.5 and its excess kurtosis 1.15×10^4 , against the TW_1 values 0.293 and 0.165 , its ratio g_V falls through 0.67 , and the Kolmogorov-Smirnov distance of the standardized sample to every candidate law grows with V , reaching 0.085 . None of this comes from the bulk.

Numerical Result 3.5. For the permutation multigraph at $V = 500,000$, the parameter-free variable s is at Kolmogorov-Smirnov distance 0.0077 from TW_1 , against 0.0227 for the legacy simple series, with scale ratio 0.9992 ± 0.0027 , and after the affine fit the distance is 0.0016 , below the typical value 0.0027 for 10^5 samples drawn from the law itself. Of the 18 samples more than eight standard deviations above $2\sqrt{3}$ across the four sizes, 17 lie within 0.002 of the three defect eigenvalues of cycle rank two computed below, and the exception, 3.5116 at $V = 50,000$, lies among the values of several defects of cycle rank three.

The moments are thus carried by a handful of far outliers, a few samples in each 10^5 , up to about 200 standard deviations of the bulk above its mean at $V = 500,000$, which inflate the sample standard deviation by about 40% and so compress the standardized bulk (Figure 3.12). The outliers are localized eigenvalues, and they are not produced by the loops and multiple edges alone. A loop contributes 2 to the diagonal, and a loop,

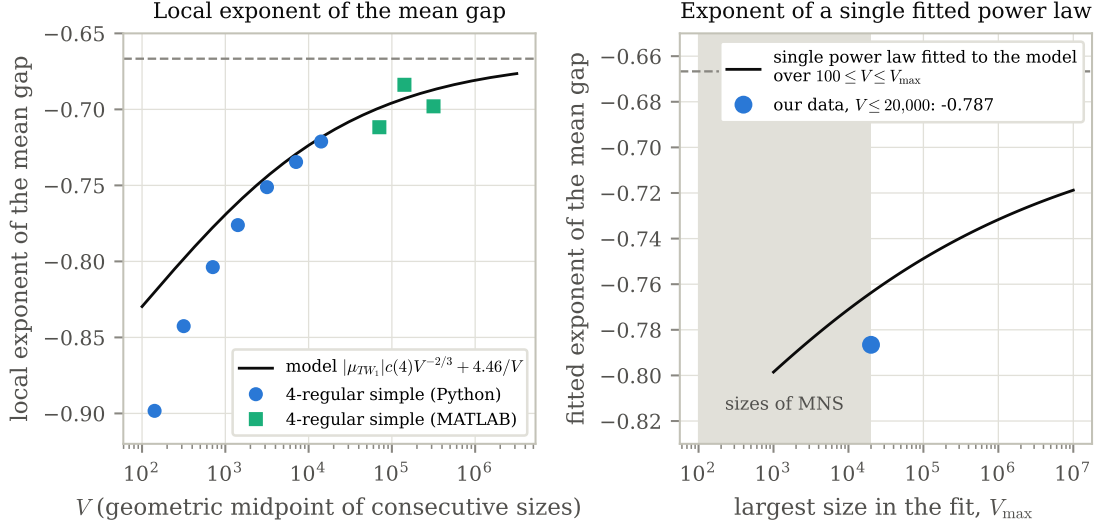


FIGURE 3.10. The origin of the 52% conjecture. Left: the local exponents of the mean gap between consecutive sizes for the two simple 4-regular series, with the prediction (3.4) of the one constant model and the limit $-2/3$ dashed. Right: the exponent reported by a single power law fitted to the model over $100 \leq V \leq V_{\max}$, against $V_{\max}$, with the value -0.787 fitted to our data over the sizes of MNS (shaded).

Local log-log slopes between consecutive sizes — a single power-law fit over small V overestimates the gap exponent

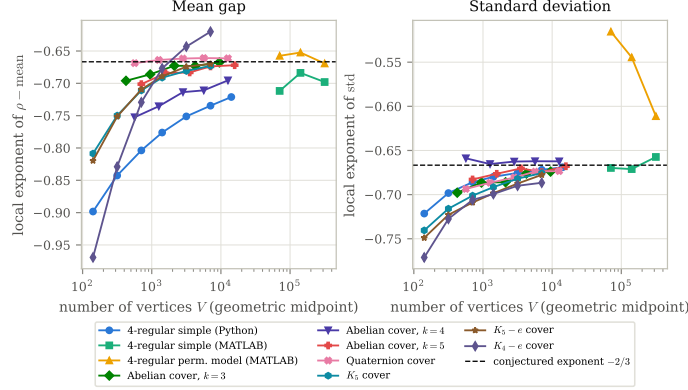


FIGURE 3.11. Local (consecutive size) log-log exponents of the mean gap and the standard deviation for all series, with the $-2/3$ line. The mean gap exponent approaches $-2/3$ from below as V grows, and so a single power law fitted over small sizes overestimates its steepness, wrongly suggesting that the gap outruns the standard deviation.

or a double edge, attached to the branches of the 4-regular tree, binds no eigenvalue above $2\sqrt{3}$, since the resulting equation $\lambda = 2 + 2y(\lambda)$, with $y(\lambda) = (\lambda - \sqrt{\lambda^2 - 12})/6$ the branch value of (2.3), has no solution above $2\sqrt{3}$, where its right hand side is $2 + 2/\sqrt{3} = 3.155$. In the language of subsection 5.2, the binding defects have cycle rank two. A triple edge, and two looped vertices joined by an edge, each bind at exactly $7/2$, a looped vertex joined to its neighbor by a double edge binds at the real root

$$\lambda = 3.5385847\dots \quad \text{of} \quad \lambda^3 = 8\lambda + 16,$$

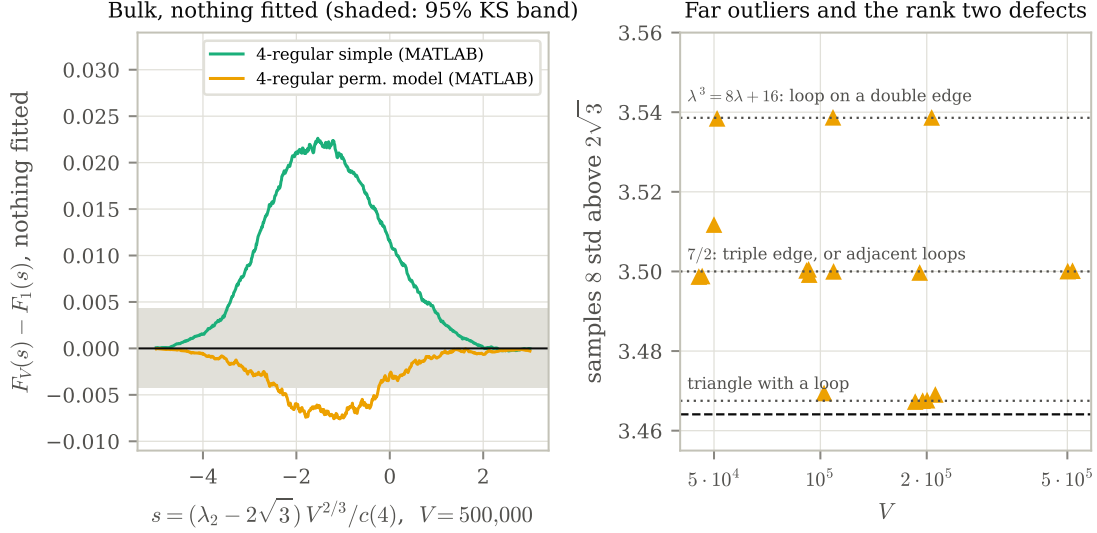


FIGURE 3.12. The permutation multigraph model. Left: $F_V(s) - F_1(s)$ for the parameter-free variable at $V = 500,000$, for the multigraph and the legacy simple series, with the 95% Kolmogorov-Smirnov band for 10^5 samples shaded: nothing is fitted, and the multigraph is closer to the law. Right: every multigraph sample more than eight standard deviations above $2\sqrt{3}$ against V , with the three computed defect eigenvalues of cycle rank two dotted.

and a triangle with a loop at one vertex binds just above the threshold, at $3.4675039 \dots$, and an exhaustive search shows that no other defect of cycle rank two on at most five vertices binds. The outliers on $7/2$ drain up to it, from 3.4985 at $V = 50,000$ to 3.4999 at $V = 500,000$. Defects of cycle rank two occur at rate $1/V$ by Proposition 5.4, and so the multigraph model has least binding rank $r^* = 2$, as does $K_4 - e$, and Corollary 5.7 applies: the defects eventually dominate the variance, growing relative to the bulk like $V^{1/3}$, which is the growth of every moment based distance observed above. Conditioning on simplicity removes all three defects, and no simple subgraph of cycle rank two on at most five vertices binds, so that the leading defect of the simple model is K_4 , which binds at $7/2$ as in the covers of K_5 but has cycle rank three and occurs at rate V^{-2} . The largest sample of the simple series at $V = 500,000$ sits only 9 standard deviations above its mean.

The lesson for experiments in this area is that statistics built from sample moments are unusable when defects of low cycle rank bind, while quantile statistics are immune to them. The simple and multigraph models share the limiting spectral distribution, the almost sure convergence of λ_2 to $2\sqrt{3}$, and the Tracy-Widom law of the bulk with its constant, and they are separated at the fluctuation scale by a deterministic shift of order $1/V$ and by a thin tail of defects of cycle rank two.

4. COVERS AND THE THREEFOLD WAY

We now turn to the main experiment. Each abelian and quaternion family isolates a definite set of isotypic blocks, and Conjecture 1.3 predicts the symmetry class of each: $\mathbb{Z}_3$ -covers should follow χ_2 , quaternion covers χ_4 , the regular graph families of section 3 χ_1 , and the mixed families $\mathbb{Z}_4$ and $\mathbb{Z}_5$ no single Tracy-Widom law at all.

4.1. Selection of the Symmetry Class. The structural result of this paper is Figure 1.1, which shows, for each family at its largest size and each candidate law, the Kolmogorov-Smirnov distance after the best affine fit to that law.

Numerical Result 4.1. After the best affine fit, every pure family is closest to the law of its own class. The simple 4-regular, K_5 , and $K_5 - e$ families are at distance 0.6 to 0.7×10^{-3} from TW_1 , against 2.1×10^{-3} or more for every other law, the $\mathbb{Z}_3$ -covers at 0.8×10^{-3} from TW_2 , against 2.6×10^{-3} for TW_4 and 3.5×10^{-3} for TW_1 , and the quaternion covers at 0.5×10^{-3} from TW_4 , against 2.7×10^{-3} for TW_2 and 5.6×10^{-3} for TW_1 , where the typical distance of 10^6 samples from their own law is 0.9×10^{-3} . In the mass to the left of the mean, the $\mathbb{Z}_3$ series converges to 0.5145 against the TW_2 value 0.515016, the quaternion series to 0.5109 against the TW_4 value 0.511069, and the regular series to 0.5201 against the TW_1 value 0.519652, with every competing class excluded by six or more binomial standard errors for the cover families and by three or more for the legacy regular sample.

The standardized statistics agree. Figure 4.2 shows that for each pure family the Kolmogorov-Smirnov distance to the standardized law of its own symmetry class falls to the sampling floor while the distance to $\widetilde{TW}_1$ (dotted) stalls. At the largest sizes, the $\mathbb{Z}_3$ family reaches distance 1.0×10^{-3} from $\widetilde{TW}_2$ while its distance to $\widetilde{TW}_1$ is stuck at 5.4×10^{-3} , and the quaternion family reaches 7.5×10^{-4} from $\widetilde{TW}_4$ with distance 8.8×10^{-3} to $\widetilde{TW}_1$. Figure 4.3 shows the mass to the left of the mean for every family.

The tails separate the classes most visibly, since the standardized Tracy-Widom laws differ most there, the left tail of TW_β decaying like $\exp(-\beta|s|^3/24)$. Figure 4.1 shows both tails of the three pure families down to probability 10^{-6} , after mapping each sample onto its own law by the affine fit and standardizing, and each follows the tail of its own class. The affine invariant moments agree as well: Figure 4.4 shows the sample skewness and excess kurtosis of each family converging to the values of its own class, (0.213, 0.093) at the largest $\mathbb{Z}_3$ size against the TW_2 values (0.224, 0.093), and (0.156, 0.042) at the largest quaternion size against the TW_4 values (0.166, 0.049), and Figure 4.5 shows the same trajectories in the plane of the two moments, where each family, the mixed ones included, moves to the point of its own limit law. Figure 4.6 gives the quantile level view of the same selection, and Figure 4.7 overlays the standardized densities of the largest datasets from five families on the $\widetilde{TW}_1$ density: at the resolution of the eye all collapse to the Tracy-Widom shape, and the discrimination between the β classes lives in the statistics above.

4.2. The Scale Constants. Conjecture 1.3 predicts the scale as well as the class, and the parameter-free variable of Definition 2.4 tests it with the constant $c(4)$ of random 4-regular graphs and the base size n as the block dimension. For the character blocks this rests on the fact that each block A_χ is the adjacency matrix of the random base graph with its entries rotated by the unimodular phases $\chi(\sigma_e)$, a perturbation which does not change the tree-like local structure or the Kesten-McKay law.

Numerical Result 4.2. At $n = 5000$ the scale ratios are 1.0083 ± 0.0006 for the $\mathbb{Z}_3$ -covers, 1.0055 ± 0.0013 for the $\mathbb{Z}_5$ -covers, and 0.9901 ± 0.0012 for the $\mathbb{Z}_4$ -covers, each converging monotonically to 1, and 1.0052 ± 0.0008 for the quaternion covers against the constant $c(4)$ with $\chi_4 = 2^{-1/6}TW_4$. Without the factor $2^{-1/6}$ the quaternion ratio would be 0.896. The location coefficients settle at the family dependent constants $a = -0.35$ for $\mathbb{Z}_3$, $+1.05$ for $\mathbb{Z}_4$, $+0.05$ for $\mathbb{Z}_5$, and -0.77 for the quaternion covers, each with standard error near 0.01.

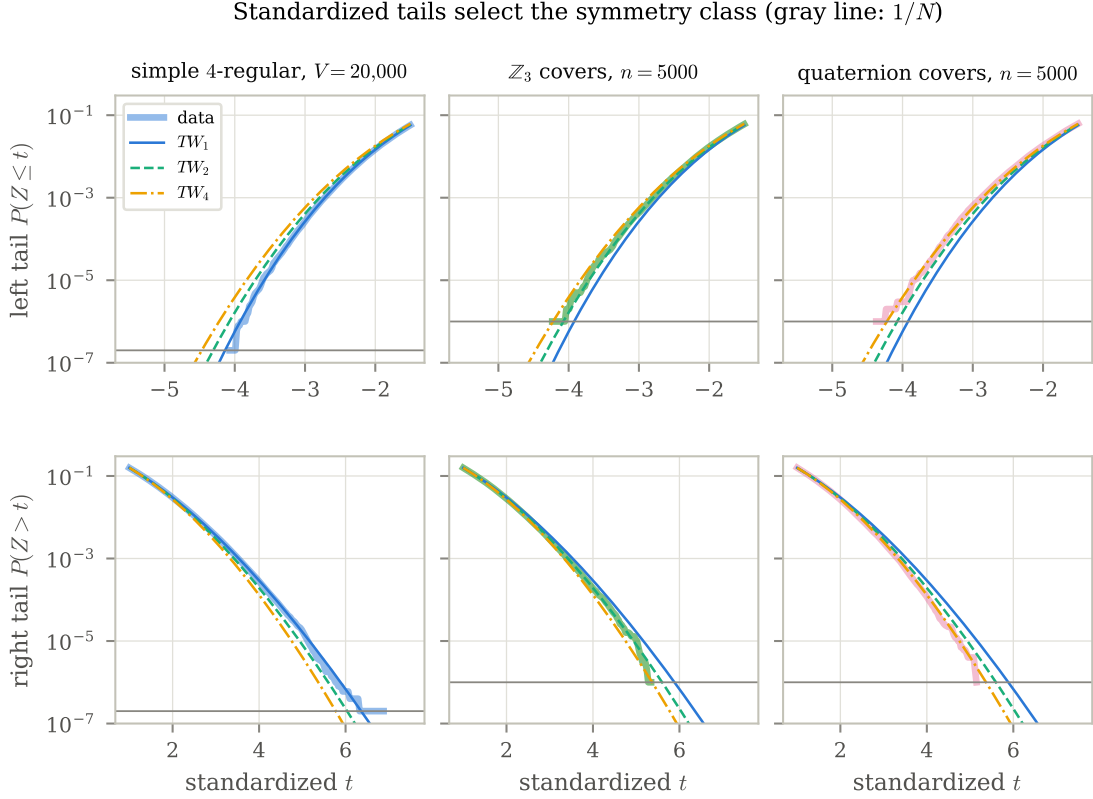


FIGURE 4.1. Left tails (top) and right tails (bottom) on logarithmic axes for the simple 4-regular graphs at $V = 20,000$, the $\mathbb{Z}_3$ -covers at $n = 5000$, and the quaternion covers at $n = 5000$, after the affine fit to the law of each family's class and standardization, against the standardized TW_1 , TW_2 , and TW_4 . The gray line is $1/N$.

The data thus selects the GSE normalization of Remark 2.3, and the quaternionic block carries the same constant as every other block. Figure 4.8 shows the parameter-free distribution functions against the conjectured laws: the differences shrink with n in every family, and after the affine fit the Kolmogorov-Smirnov distances at the largest size are between 0.5×10^{-3} and 1.2×10^{-3} , at the sampling floor. The location coefficients of the covers are all much smaller than the 4.47 of the simple graphs (Figure 3.7 and Table 4.2).

4.3. The Mixed Families and the Maximum Laws. The largest new eigenvalue of a $\mathbb{Z}_4$ -cover is the maximum of a block maximum of class $\beta = 1$ and one of class $\beta = 2$, and that of a $\mathbb{Z}_5$ -cover is the maximum of two block maxima of class $\beta = 2$, and so under Conjectures 1.3 and 1.4 neither should converge to any single Tracy-Widom law. This is visible in the raw statistics: the Kolmogorov-Smirnov distance of the standardized $\mathbb{Z}_4$ family to every candidate law grows with V , reaching 1.9×10^{-2} against $\widetilde{TW}_1$ at the largest size, and the $\mathbb{Z}_5$ family, nearest $\widetilde{TW}_1$ at moderate sizes, drifts away from it as V grows. If the two blocks of a family fluctuate independently with the same centering $2\sqrt{d-1}$ and the same scale constant, as Conjecture 1.3 asserts, then the largest new eigenvalue converges, on the scale $c(d)n^{-2/3}$, to

$$M = \max(X_1, X_2),$$

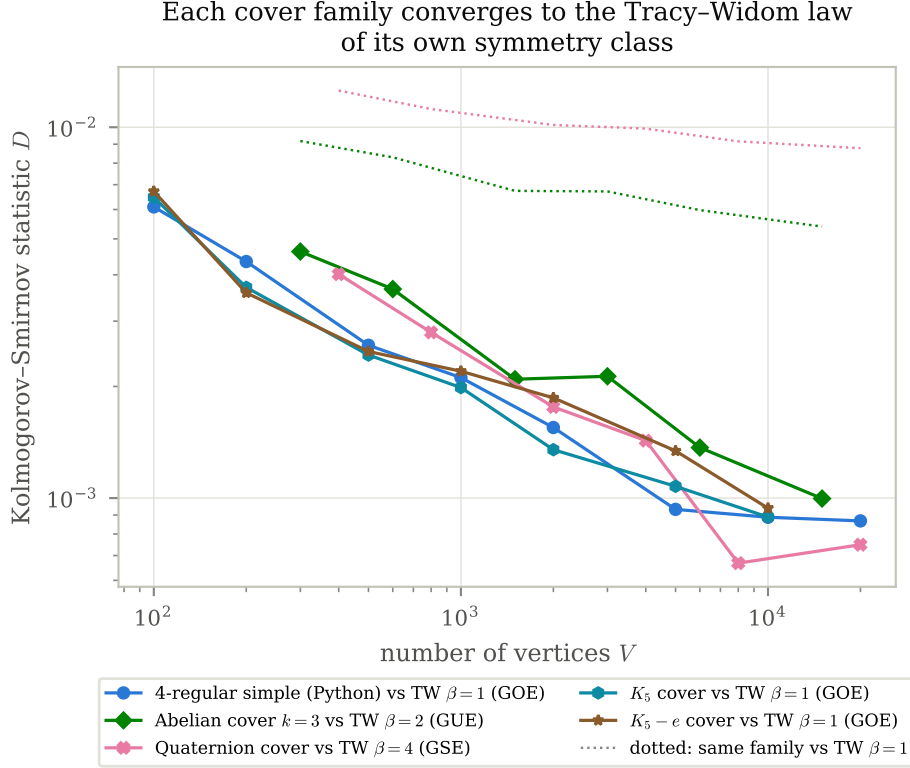


FIGURE 4.2. Kolmogorov-Smirnov distance of each pure family to the standardized Tracy-Widom law of its own symmetry class (solid): regular graphs and the fixed base K_5 and $K_5 - e$ cover families to $\beta = 1$, $\mathbb{Z}_3$ -covers to $\beta = 2$, quaternion covers to $\beta = 4$, with the distance to $\beta = 1$ dotted for contrast. The matching class distances fall to the sampling noise floor. The mixed families $\mathbb{Z}_4$ and $\mathbb{Z}_5$ match no single law.

where X_1, X_2 are independent Tracy-Widom variables of the appropriate classes, with distribution function $F_1 F_2$ for $\mathbb{Z}_4$ and F_2^2 for $\mathbb{Z}_5$.

Numerical Result 4.3. After the affine fit, the Kolmogorov-Smirnov distance of each mixed family to its maximum law falls with n , to 1.2×10^{-3} for $\mathbb{Z}_4$ and 0.7×10^{-3} for $\mathbb{Z}_5$ at $n = 5000$, while the distances to the three Tracy-Widom laws grow. The $\mathbb{Z}_5$ family matches the mass left of the mean and the Ramanujan probability of its maximum law to three decimal places at $n = 5000$, and the ratio g_V to within 3×10^{-3} . With its location coefficient $a = 1.05$, the finite size law (2.6) predicts the $\mathbb{Z}_4$ Ramanujan probability $F_1 F_2(a/(c(4)n^{1/3})) = 0.82291$ at $n = 5000$, against the observed 0.82293.

The drift away from the pure laws is invisible at moderate sizes: up to $n = 2000$ the $\mathbb{Z}_5$ family is closer in shape to TW_1 than to its own maximum law, whose skewness and kurtosis are close to those of TW_1 , and with 10^6 samples at $V = 2500$ the Kolmogorov-Smirnov test accepts $\widetilde{TW}_1$ ($p = 0.81$), a caution for experiments at the scale of MNS (Figure 4.9). The $\mathbb{Z}_4$ family is still drifting at $n = 5000$, in each statistic monotonically toward the predicted value, and its location coefficient, the largest among the covers, is the cause. The maximum structure is visible at the moment level as well: at $n = 5000$ the $\mathbb{Z}_4$ skewness and excess kurtosis are $(0.570, 0.658)$ against the $\max(TW_1, TW_2)$

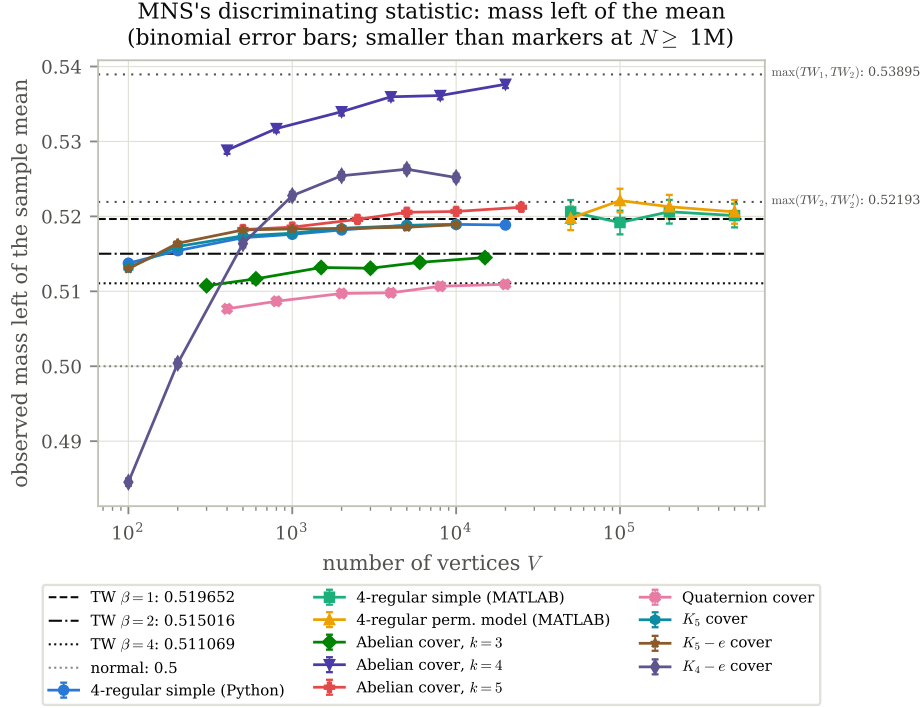


FIGURE 4.3. The fraction of samples below the sample mean against V for all families, with the Tracy-Widom $\beta = 1, 2, 4$ and normal reference values. This replicates MNS's decisive experiment at fifty times their sample size: with 10^6 to 5×10^6 samples the binomial standard error is between 2×10^{-4} and 5×10^{-4} and the four candidate laws are separated by tens of standard errors. Each pure family selects the value of its own symmetry class, while the mixed family $\mathbb{Z}_4$ drifts to ≈ 0.538 , consistent with the maximum law of Table 4.1 and no Tracy-Widom law.

Third and fourth moments converge to the Tracy-Widom values of each family's symmetry class (permutation model omitted: kurtosis up to $\sim 10^4$)

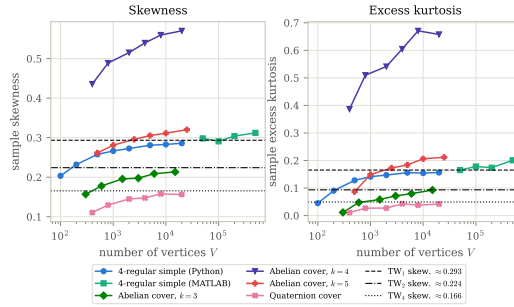


FIGURE 4.4. Sample skewness and excess kurtosis against V with the Tracy-Widom $\beta = 1, 2, 4$ reference values. The regular families converge to the $\beta = 1$ moments, $\mathbb{Z}_3$ to $\beta = 2$, and quaternion to $\beta = 4$. The multigraph model (excess kurtosis up to 10^4) and the fixed base cover families, whose third and fourth moments are dominated by the far outlier tails of section 5, are omitted.

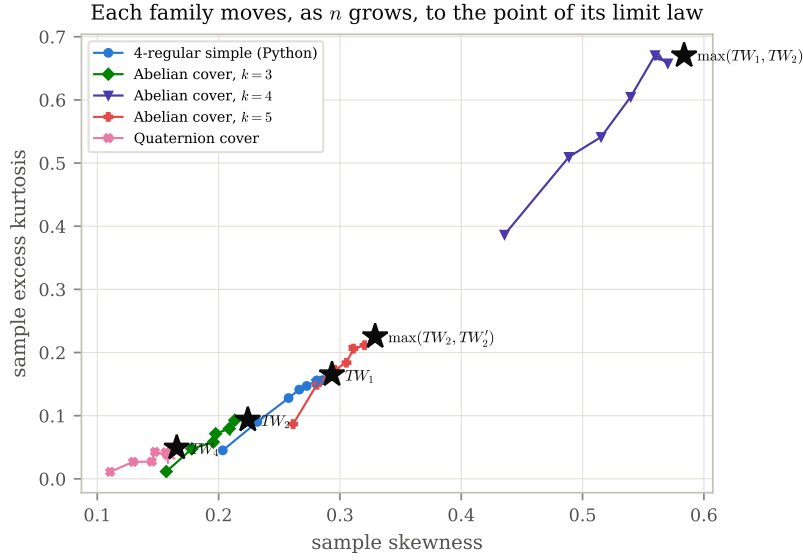


FIGURE 4.5. Sample skewness against excess kurtosis for the simple 4-regular, cyclic, and quaternion families, one marker per size, with the arrow at the largest size. Each family moves to the point of its limit law (stars): TW_1 , TW_2 , TW_4 , or the maximum laws of subsection 4.3.

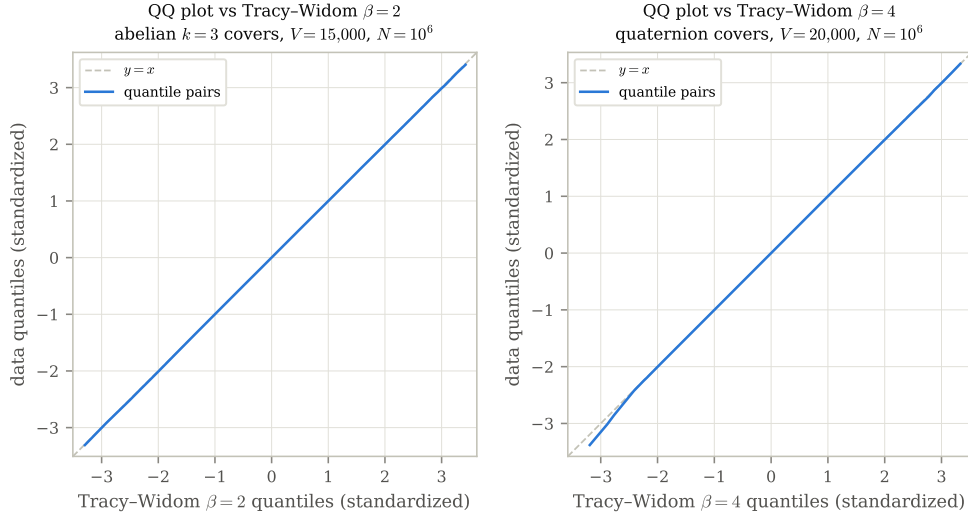


FIGURE 4.6. Quantile-quantile plots of the largest $\mathbb{Z}_3$ cover dataset against Tracy-Widom $\beta = 2$ (left) and of the largest quaternion cover dataset against $\beta = 4$ (right), both standardized, probability range 10^{-4} to 0.999.

values $(0.584, 0.670)$, and the $\mathbb{Z}_5$ values are $(0.320, 0.212)$ against $(0.329, 0.225)$, in each case drifting monotonically toward the prediction (Figure 4.5).

4.4. The Ramanujan Probabilities. Computing the distribution functions $F_1 F_2$ and F_2^2 numerically gives the predicted constants in Table 4.1: the ratio $|\mu_M|/\sigma_M$, the mass of M left of its mean, and the Ramanujan probability $F_{\beta_1}(0)F_{\beta_2}(0)$. Table 4.1 is the experimental form of Conjecture 1.4, and Table 4.2 lists the finite size parameters of every family.

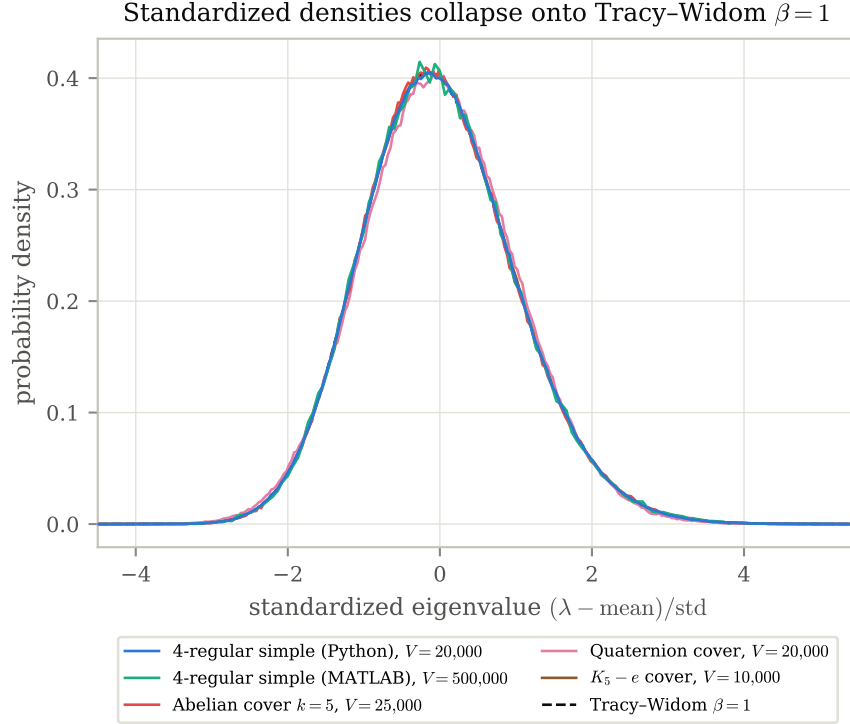


FIGURE 4.7. Standardized empirical densities of the largest datasets from five families overlaid on the standardized $\beta = 1$ Tracy-Widom density. The β classes are indistinguishable at this resolution, and are separated by the statistics of Figures 1.1, 4.1, 4.2, 4.3, and 4.4.

Numerical Result 4.4. Each family approaches the product $\prod F_\beta(0)$ over its blocks: one block of class $\beta = 2$ gives $F_2(0) = 0.96937\dots$ for $\mathbb{Z}_3$, against the observed 0.96793, two give $F_2(0)^2 = 0.93968\dots$ for $\mathbb{Z}_5$, against 0.94011, the mixed $\mathbb{Z}_4$ product $F_1(0)F_2(0) = 0.80643\dots$ is approached from above through 0.82293, and the quaternionic block gives $F_4(0) = 0.99857\dots$ against 0.99825, all at $n = 5000$. Evaluating each law at its shifted point $a/(cN^{1/3})$ reproduces every observed Ramanujan probability in Table 4.1 to within 3×10^{-4} , and to within 6×10^{-4} for $K_4 - e$.

Figure 4.10 shows the full curves, and Figure 4.11 the underlying ratios g_V , each family flattening onto the ratio of its own limit law. In particular the limiting probability of being Ramanujan is not one number but a representation theoretic invariant of the family, always strictly between 0 and 1.

5. COVERS OF A FIXED BASE GRAPH

We now fix the base graph and let only the cover vary, testing three bases: the complete graph K_5 , whose covers are 4-regular and calibrate the fixed base model against section 3, the irregular graph $K_5 - e$, and the cycle rank two graph $K_4 - e$ of subsection 5.3. Nothing in Conjecture 1.6 requires the base graph to be regular, and in the irregular case nothing is proven: the largest new eigenvalue should concentrate at the spectral radius $\rho(B)$ of the universal cover, an algebraic number lying between $2\sqrt{\delta - 1}$ and $2\sqrt{\Delta - 1}$, where δ and Δ are the minimum and maximum degrees of B , with TW_1 fluctuations on the scale $(C_B n k)^{-2/3}$ and limiting Ramanujan probability $F_1(0)$, universally in the base graph. The threshold $\rho(B)$ is also the benchmark of the

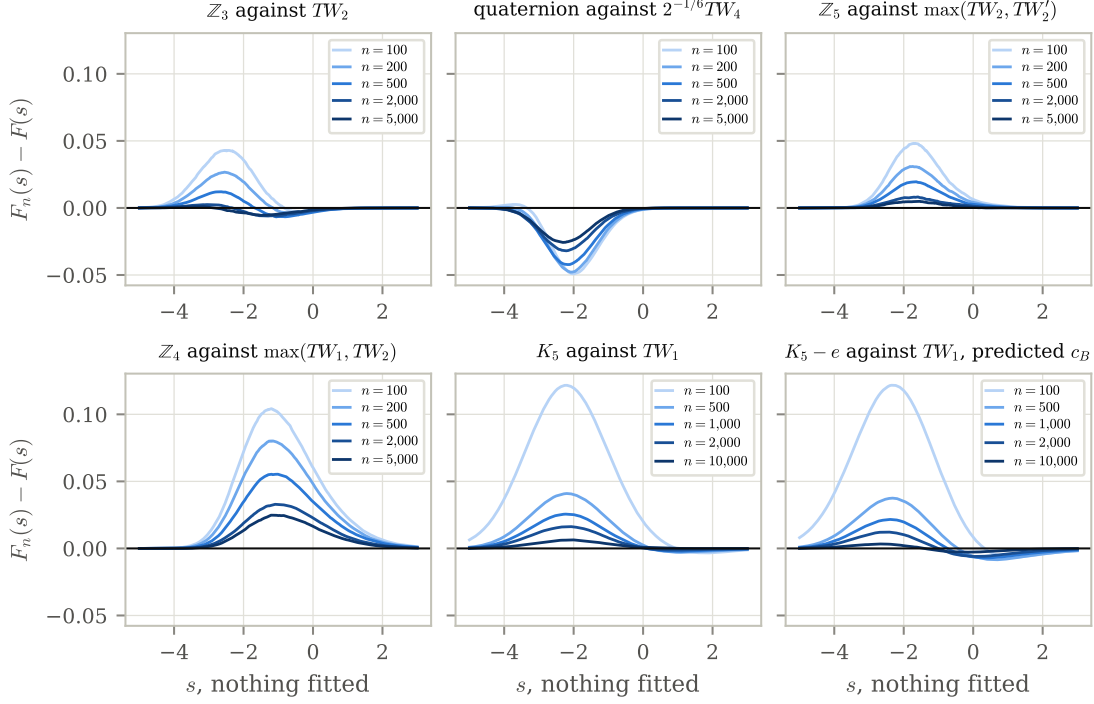


FIGURE 4.8. Parameter-free comparison of six cover families with their conjectured laws: $F_n(s) - F(s)$, where F_n is the empirical distribution function of $s = (\lambda - \rho)N^{2/3}/c$ with the predicted c , at five sizes each (lightest: smallest). The $K_5 - e$ panel uses the universal cover constant $C_B^{-2/3} = 0.695228$ of (2.4). Nothing is fitted.

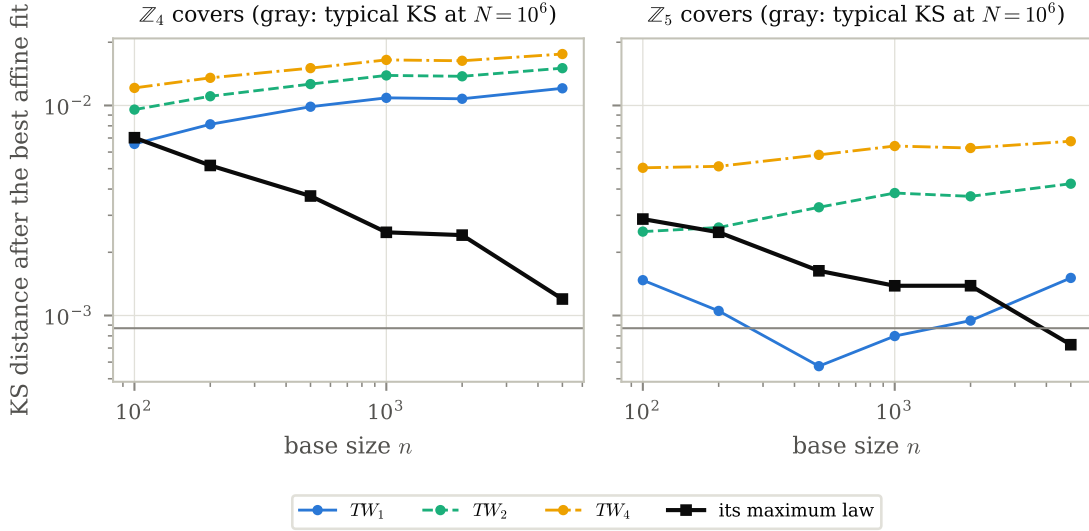


FIGURE 4.9. Kolmogorov-Smirnov distance after the best affine fit, for the $\mathbb{Z}_4$ (left) and $\mathbb{Z}_5$ (right) cover families, to the three Tracy-Widom laws and to the family's own maximum law, $\max(TW_1, TW_2)$ and $\max(TW_2, TW_2')$. Only the distance to the maximum law falls with n .

family	limit law	g_V	mass left of mean	$P(\text{Ramanujan})$
simple 4-regular	TW_1	0.9993 (0.9515)	0.52011 (0.51965)	0.84352 (0.83191)
$\mathbb{Z}_3$ -covers	TW_2	1.9369 (1.9640)	0.51450 (0.51502)	0.96793 (0.96937)
$\mathbb{Z}_4$ -covers	$\max(TW_1, TW_2)$	0.8811 (0.8099)	0.53763 (0.53895)	0.82293 (0.80643)
$\mathbb{Z}_5$ -covers	$\max(TW_2, TW'_2)$	1.6245 (1.6219)	0.52122 (0.52193)	0.94011 (0.93968)
quaternion covers	TW_4	3.1263 (3.2061)	0.51094 (0.51107)	0.99825 (0.99857)
K_5 covers	TW_1	0.9551 (0.9515)	0.51891 (0.51965)	0.83281 (0.83191)
$K_5 - e$ covers	TW_1	0.9387 (0.9515)	0.51890 (0.51965)	0.82916 (0.83191)
$K_4 - e$ covers	TW_1	0.7326 (0.9515)	0.52518 (0.51965)	0.79490 (0.83191)

TABLE 4.1. Observed statistics at the largest available size of each family against the values predicted by Conjectures 1.3, 1.4, and 1.6: the ratio $g_V = (\rho - \mu_V) / \sigma_V$ of the mean gap to the standard deviation, the mass left of the sample mean, and the fraction of samples below the Ramanujan threshold ρ ($\rho = 2\sqrt{3}$ except for the $K_5 - e$ and $K_4 - e$ rows, where ρ is the spectral radius of the universal cover of the base). Predicted values for the maximum laws were computed numerically from the Tracy-Widom distributions assuming independent blocks on a common scale. Each cell shows the observed value with the predicted limit in parentheses. Observed values are at $V = 500,000$ (simple), $n = 5000$ (abelian and quaternion covers), and $V = 10,000$ (the fixed base families). The remaining gaps in the Ramanujan column are the $1/n$ shifts of Table 4.2.

family	size	predicted scale	scale ratio γ	location a	KS after affine fit
simple 4-regular	$V = 20,000$	$c(4) = 0.832683$	1.0031 ± 0.0003	4.453 ± 0.020	0.7×10^{-3}
simple (legacy)	$V = 500,000$	$c(4)$	1.0036 ± 0.0033	4.30 ± 0.37	1.2×10^{-3}
multigraph	$V = 500,000$	$c(4)$	0.9992 ± 0.0027	-1.31 ± 0.34	1.6×10^{-3}
$\mathbb{Z}_3$ -covers	$n = 5000$	$c(4)$	1.0083 ± 0.0006	-0.351 ± 0.018	0.8×10^{-3}
$\mathbb{Z}_4$ -covers	$n = 5000$	$c(4)$	0.9901 ± 0.0012	1.022 ± 0.026	1.2×10^{-3}
$\mathbb{Z}_5$ -covers	$n = 5000$	$c(4)$	1.0055 ± 0.0013	0.022 ± 0.028	0.7×10^{-3}
quaternion covers	$n = 5000$	$c(4)$	1.0052 ± 0.0008	-0.742 ± 0.026	0.5×10^{-3}
K_5 covers	$V = 10,000$	$c(4)$	1.0091 ± 0.0003	0.073 ± 0.012	0.7×10^{-3}
$K_5 - e$ covers	$V = 10,000$	$C_B^{-2/3} = 0.695228$	1.0119 ± 0.0003	-0.233 ± 0.014	0.6×10^{-3}
$K_4 - e$ covers	$V = 10,000$	$C_B^{-2/3} = 0.212519$	1.0441 ± 0.0009	-0.922 ± 0.007	2.8×10^{-3}

TABLE 4.2. The finite size law (2.6) at the largest size of each family: the predicted scale, the scale ratio, the location coefficient, and the Kolmogorov-Smirnov distance of the sample to its conjectured law after the affine fit, with standard errors from ten batches. The typical distance of a sample from its own law is 0.4×10^{-3} at 5×10^6 samples, 0.9×10^{-3} at 10^6 , and 2.7×10^{-3} at 10^5 .

X-Ramanujan property of Mohanty and O'Donnell [85], who construct infinite families of finite graphs whose second eigenvalue is at most the spectral radius of a fixed infinite cover, and Hoory [56] bounds $\rho(B)$ below by $2\sqrt{\bar{d}} - 1$, where $\bar{d}$ is the average degree of B .

The main base is $K_5 - e$, the complete graph on five vertices minus an edge, one of the smallest irregular base graphs of minimum degree three, with eigenvalues $1 + \sqrt{7} = 3.6457\dots$, 0 , -1 , -1 , and $1 - \sqrt{7}$. Since the largest old eigenvalue exceeds $\rho(K_5 - e)$, the largest new eigenvalue of a cover is not even its second largest eigenvalue, so that the exact removal of the lifted base spectrum matters. By a result of McKay [82],

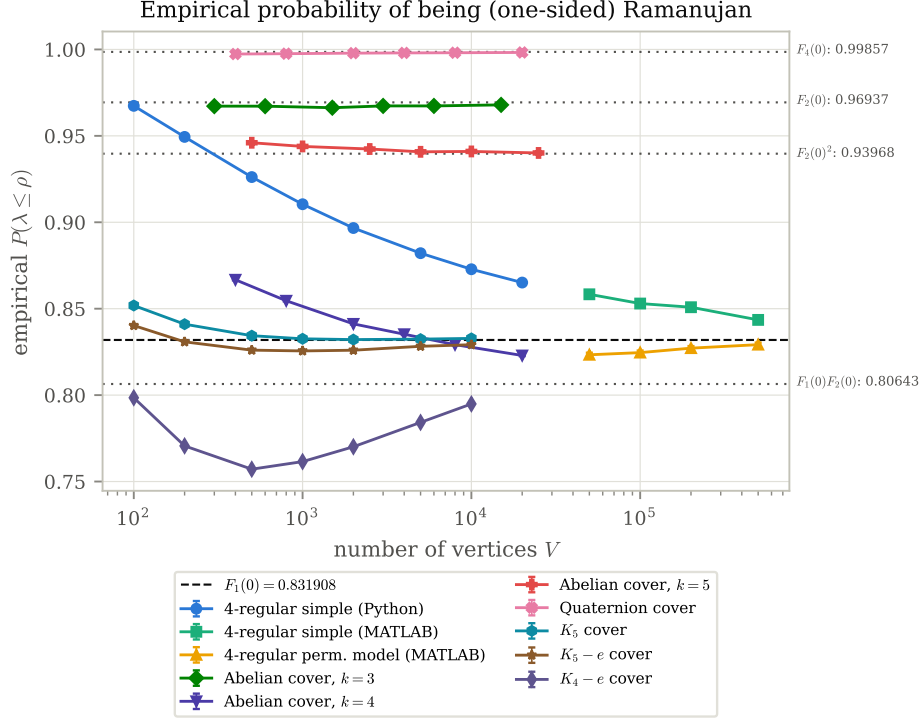


FIGURE 4.10. Empirical $P(\lambda \leq \rho)$ against V for all families, where ρ is the family's Ramanujan threshold, with the limit $F_1(0) = 0.8319081\dots$ for the $\beta = 1$ families. Each cover family approaches the product of Tracy-Widom values of Conjecture 1.4 listed in Table 4.1.

$\rho(K_5 - e)$ is the largest zero of

$$x^{14} - 20x^{12} + 122x^{10} - 152x^8 - 1295x^6 + 4540x^4 - 5948x^2 - 2000,$$

an irreducible polynomial whose roots admit no expression in radicals, giving

$$\rho(K_5 - e) = 3.26287646593635862827\dots,$$

which we re-derived and verified independently, and which also follows from the variational formula of Garza-Vargas and Kulkarni [46]. Note the contrast with the regular case, where $\rho = 2\sqrt{d-1}$: for a general base even the centering of the conjectured law is a non-trivial algebraic quantity (see [12, 46, 82] on computing ρ), and so is the scale constant C_B of (2.4).

5.1. The Bulk Law and the Amplitudes. We sampled the K_5 and $K_5 - e$ families at the scale of the strongest regular graph data: 5×10^6 independent uniform covers per size, at cover degrees $20 \leq k \leq 2000$, giving graphs on $V = 5k$ between 100 and 10,000 vertices. The covers of K_5 are 4-regular graphs with Ramanujan threshold $2\sqrt{3}$, of a different fine structure than the permutation model, since the vertex set carries a fixed partition into five independent fibers with edges only between fibers adjacent in K_5 , and so they test whether the limit law and its constants depend on anything beyond the degree. The covers of $K_5 - e$ test Conjecture 1.6 where no theorem reaches, around the algebraic centering $\rho(K_5 - e)$ and on the scale set by the new constant $C_{K_5 - e}$.

Numerical Result 5.1. For the covers of K_5 and of $K_5 - e$, the Kolmogorov-Smirnov distance of the standardized sample to $\widetilde{TW}_1$ falls monotonically with V , to 8.9×10^{-4}

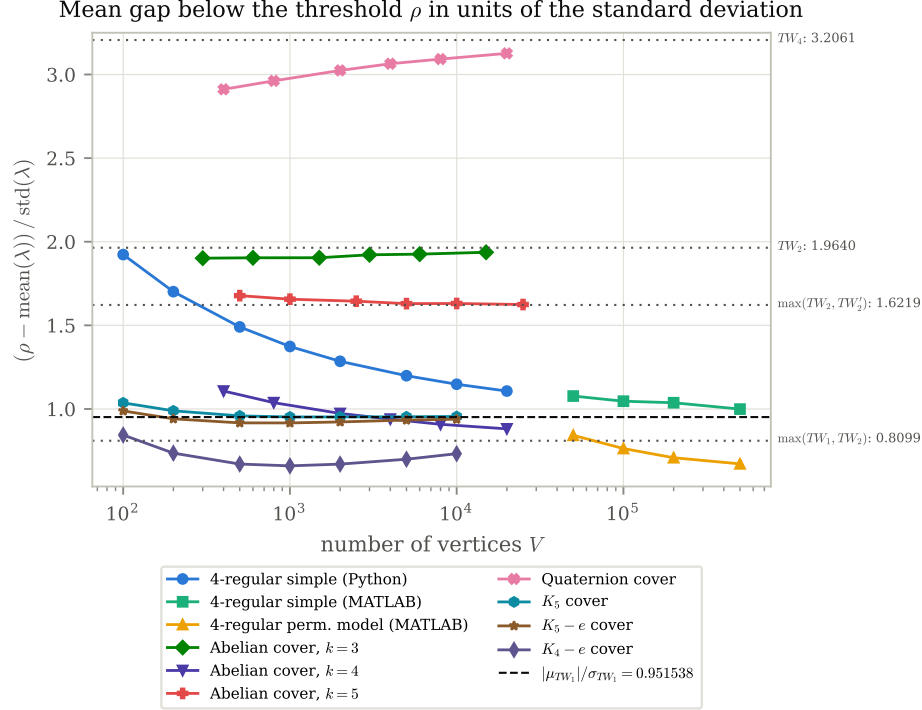


FIGURE 4.11. The ratio $g_V = (\rho - \mu_V) / \sigma_V$ against V for all families, where ρ is the family's Ramanujan threshold, with the $\beta = 1$ reference $|\mu_{TW_1}| / \sigma_{TW_1} = 0.951537\dots$ dashed. MNS conjectured this ratio tends to 0, however each family plateaus at the ratio of its own limit law: 1.96 for $\mathbb{Z}_3$, 1.62 for $\mathbb{Z}_5$, 3.21 for quaternion covers, and the $\beta = 1$ reference for the regular and fixed base families.

and 9.4×10^{-4} at $V = 10,000$, equal to the simple 4-regular family's distance at the same size, while the distances to the other laws stall or grow, ending at 4.0×10^{-3} ($\widetilde{TW}_2$), 7.9×10^{-3} ($\widetilde{TW}_4$), and 1.9×10^{-2} (normal) for $K_5 - e$. The mass left of the mean reaches 0.51891 and 0.51890, against the TW_1 value 0.519652, the ratio g_V reaches 0.9551 and 0.9387, and the empirical Ramanujan probabilities at $V = 10,000$ are

$$P(\lambda_{\text{new}} \leq 2\sqrt{3}) = 0.83281, \quad P(\lambda_{\text{new}} \leq \rho(K_5 - e)) = 0.82916,$$

against the conjectured $F_1(0) = 0.8319081\dots$

Every statistic selects the $\beta = 1$ class, as the reality of the standard representation predicts, and for $K_5 - e$ it does so around a centering which is not $2\sqrt{d-1}$ for any d (Figures 5.1, 5.2, and 5.3). A two sample comparison is sharper still: the Kolmogorov-Smirnov distance between the standardized simple and K_5 samples at $V = 10,000$ is 3.7×10^{-4} ($p = 0.88$), while each lies 8.9×10^{-4} from $\widetilde{TW}_1$ ($p \approx 10^{-3}$), and so at 10^7 combined samples the two models are statistically indistinguishable from one another in shape while both still differ measurably from the limit. At small sizes the families are well separated ($p = 2 \times 10^{-15}$ at $V = 100$), merging from $V \approx 5000$ on. The local exponents of the mean gap and standard deviation descend to -0.673 and -0.676 for K_5 and -0.670 and -0.678 for $K_5 - e$ over the last pair of sizes, approaching $-2/3$ as for the regular families (Figure 3.11).

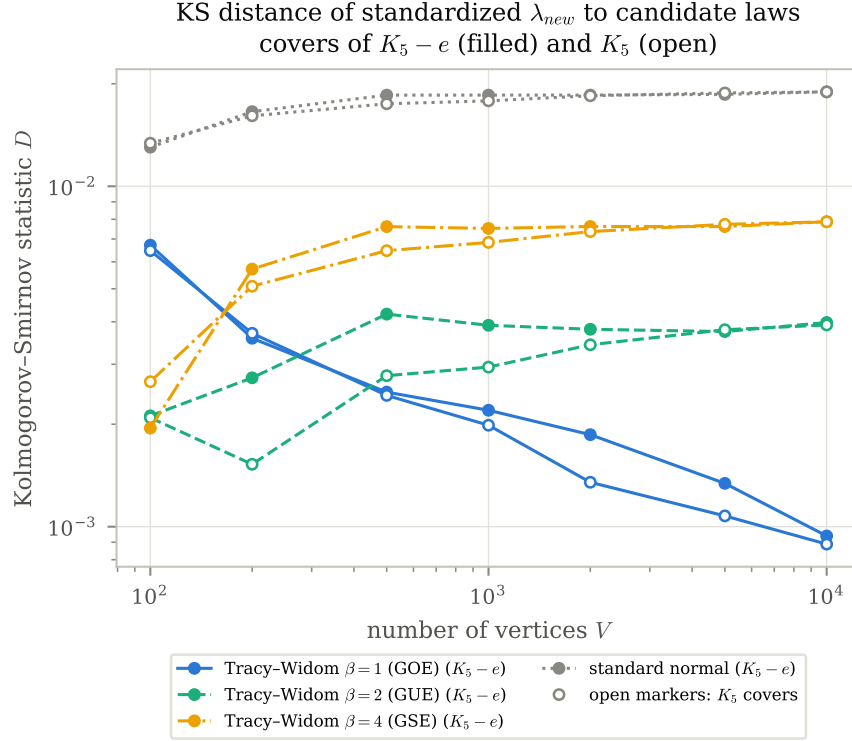


FIGURE 5.1. Kolmogorov-Smirnov distance of the standardized largest new eigenvalue to the four candidate laws for covers of $K_5 - e$ (filled markers, standardized around $\rho(K_5 - e)$) and of K_5 (open markers), 5×10^6 samples per size. For both families only the distance to $\beta = 1$ keeps falling.

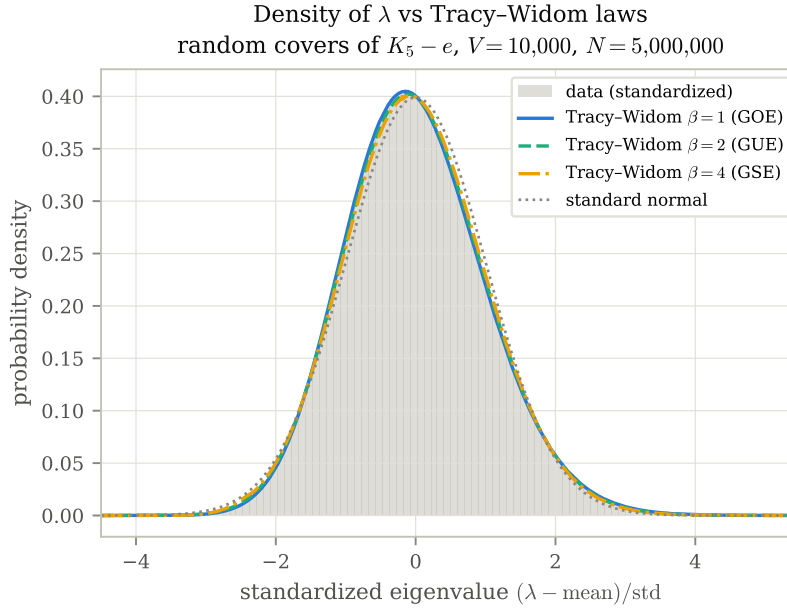


FIGURE 5.2. Standardized empirical density of the largest new eigenvalue over 5×10^6 random covers of $K_5 - e$ at $V = 10,000$, against the standardized Tracy-Widom densities and the standard normal.

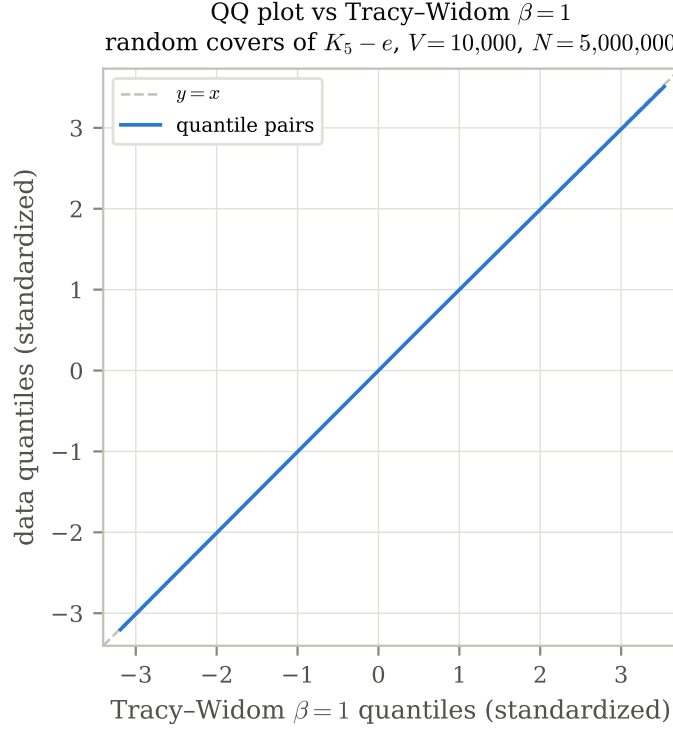


FIGURE 5.3. Quantile-quantile plot of the largest $K_5 - e$ dataset against $\widetilde{TW}_1$, probability range 10^{-4} to 0.999: the bulk of the law around the algebraic centering $\rho(K_5 - e)$ is Tracy-Widom at quantile resolution.

The amplitudes are the sharpest test of Conjecture 1.6, since for $K_5 - e$ the predicted constant $C_B^{-2/3} = 0.695228 \dots$ of (2.4) was computed from the universal cover alone, with no reference to the data.

Numerical Result 5.2. For the covers of $K_5 - e$, the fitted amplitudes $\gamma_V c(4)$ descend through 0.7209, 0.7085, and 0.7035 at $V = 2000$, 5000, and 10,000, and a fit $c + bV^{-2/3}$ over $V \geq 1000$ or $V \geq 2000$ extrapolates to 0.6943 to 0.6949, within 0.15% of the prediction 0.695228. The same fits for K_5 give 0.8315 to 0.8319, against $c(4) = 0.832683$.

The amplitude constant for covers of K_5 is thus the regular graph constant $c(4)$ itself, in the convention with V the cover size, as the Kesten-McKay law of μ_{K_5} requires (Figure 5.4). The speed of convergence differs, however: the K_5 family sits within 10^{-3} of the limiting Ramanujan probability from $V = 1000$ onward (0.83258 at $V = 1000$), while the simple 4-regular series is still at 0.87282 at $V = 10,000$, and the K_5 ratio g_V is already 0.9522 at $V = 1000$ against the simple series' 1.148 at ten times that size. The difference lies in the location coefficient, which is $a \approx 0.03$ for K_5 against 4.47 for the simple graphs (Table 4.2), and not in the scale, which for K_5 converges if anything more slowly, with scale ratio 1.046 at $V = 1000$ against 1.027. The Ramanujan probability does not see the scale, only the shift.

5.2. Localized Defects and the Far Tails. The third and fourth moments tell a different story, and the accounting is instructive. The excess kurtosis of the $K_5 - e$ family grows with V , from 0.26 at $V = 100$ to 2.52 at $V = 10,000$ against the TW_1 value 0.165, and its skewness drifts upward through 0.361 against 0.293, while the same statistics for K_5 are 0.28 and 0.292. Figure 5.7 locates the cause in a thin tail of far

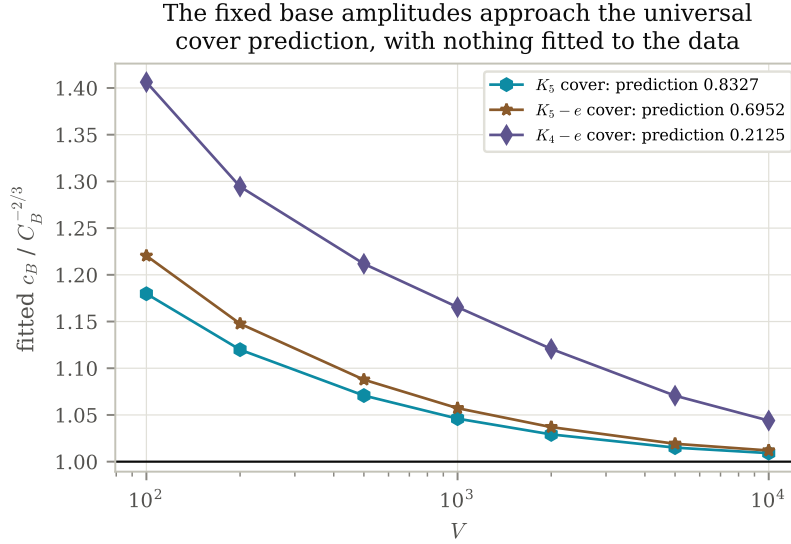


FIGURE 5.4. The fitted amplitudes of the three fixed base families over the universal cover predictions $C_B^{-2/3}$ of (2.4), none of which was fitted to the data. K_5 and $K_5 - e$ converge to 1 together, while $K_4 - e$, whose square root regime is narrow, converges more slowly.

outliers, at frequencies near 10^{-5} : at $V = 10,000$ a fraction 9×10^{-6} of the $K_5 - e$ samples lie above $\rho + 10\sigma_V$, while the extreme sample of the simple 4-regular family sits only 9 standard deviations out. These outliers are the tangles of [38] made visible sample by sample, and they admit a complete quantitative description, one half of which we can prove. For regular bases, Addario-Berry and Griffiths [1] showed that large new eigenvalues of random lifts are caused by small induced subgraphs, up to a constant factor, and the propositions below make this exact for the rate and the location of each defect.

Note that every subgraph of a cover projects to the base graph by a locally injective map, since distinct neighbors of a vertex of a cover lie over distinct neighbors of its projection. With this in mind, we make the following definition.

Definition 5.3. Let H be a connected graph of minimum degree at least two, with cycle rank $r(H) = e(H) - v(H) + 1$, and let $\varphi : H \rightarrow B$ be locally injective, an *immersion*. Let H^* be the infinite graph obtained from H by attaching, at each vertex x of H and each neighbor of $\varphi(x)$ not used by an edge of H at x , the corresponding branch of the universal cover of B . The *defect eigenvalue* of H is

$$\lambda_B(H) = \|A_{H^*}\|,$$

the spectral norm of the adjacency operator of H^* , maximized over the immersions φ when there are several, and we say that H *binds* if $\lambda_B(H) > \rho(B)$.

When B is a bouquet, H^* is the tree extension used by Dong and McKenzie [33] in their proof that every sequence $d \geq \mu_2 \geq \dots \geq \mu_k \geq 2\sqrt{d-1}$ is a limit of top eigenvalues of d -regular graphs. Deep inside any branch, H^* contains arbitrarily large balls of the universal cover, and so $\lambda_B(H) \geq \rho(B)$ always. When H binds, $\lambda_B(H)$ lies above the essential spectrum, since removing finitely many vertices from H^* leaves a union of branches, and $\lambda_B(H)$ is computable to any precision from the branch values (2.3) at real $\lambda > \rho(B)$: each attached branch contributes its value to the diagonal,

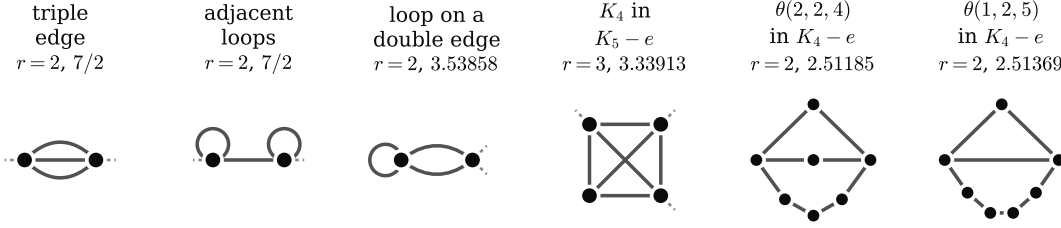


FIGURE 5.5. The leading defects, with their cycle ranks r and defect eigenvalues $\lambda_B(H)$: the three binding defects of cycle rank two of the permutation multigraph (subsection 3.4), K_4 immersed in $K_5 - e$ (in K_5 it has a fourth attached branch and binds at $7/2$), and the two shortest binding generalized thetas of $K_4 - e$ (subsection 5.3, branches not drawn). Dashed stubs are attached branches of the universal cover.

and $\lambda_B(H)$ is the largest λ with $\lambda_{\max}(A_H + D(\lambda)) = \lambda$, where $D(\lambda)$ is the diagonal matrix of attached branch values (for eigenvalues of graphs with attached infinite tails see also [47]). Figure 5.5 shows the leading defects of our families. Two elementary facts organize the far tails: copies of H appear in a random cover at a polynomial rate governed by the cycle rank, and any copy forces a new eigenvalue near $\lambda_B(H)$.

Proposition 5.4. *Let B be a connected base graph, let H be a connected graph of minimum degree at least two admitting a locally injective map to B , and let X_H be the number of subgraphs of a uniformly random degree k cover H_k of B isomorphic to H . Then*

$$E[X_H] = c_H k^{1-r(H)} (1 + O(k^{-1})), \quad c_H = \frac{J(H, B)}{|\text{Aut}(H)|},$$

where $J(H, B)$ is the number of locally injective homomorphisms from H to B , and, when $r(H) \geq 2$,

$$P(X_H \geq 1) = c_H k^{1-r(H)} (1 + o(1)).$$

Proof. Suppose first that B is simple, as are all of our fixed bases, since loops and multiple edges require only notational changes. Copies of H in the cover correspond to pairs consisting of a locally injective homomorphism $\psi : H \rightarrow B$ and a lift of ψ , with each copy counted $|\text{Aut}(H)|$ times. Fix ψ , and choose for each vertex of H a lift in the fiber over its image, injectively within each fiber, which can be done in $k^{v(H)} (1 + O(k^{-1}))$ ways. The chosen vertices span a copy of H exactly when, for each edge e of B , the permutation σ_e maps every specified source to its specified target. If m_e edges of H lie over e , then the sources are distinct and the targets are distinct, by fiber injectivity and the local injectivity of ψ , and so this event has probability $(k - m_e)!/k! = k^{-m_e} (1 + O(k^{-1}))$, independently across the edges of B . Multiplying and summing over ψ gives the expectation, since $v(H) - e(H) = 1 - r(H)$.

For the probability, we have that $E[X_H^2] = E[X_H] + E[Y]$, where Y counts ordered pairs of distinct copies. The union of two distinct copies is either disconnected, and such pairs contribute $c_H^2 k^{2-2r} (1 + O(k^{-1}))$ by the computation above, or connected of cycle rank at least $r+1$: if two distinct copies overlap, their intersection is a proper subgraph F of H , and $r(U) = 2r(H) - 1 - r(F) + q$ for the union U , where q is the number of components of F and $r(F) = e(F) - v(F) + q$, so it suffices that $r(F) \leq r(H) - 2 + q$. For connected F , with $q = 1$, this says $r(F) \leq r(H) - 1$, which holds since a proper connected subgraph of a graph of minimum degree two has strictly smaller cycle rank:

a path leaving the subgraph cannot terminate, and closes an independent cycle whether it returns to the subgraph or to itself. For disconnected F , with $q \geq 2$, it follows from $r(F) \leq r(H)$, since the cycle spaces of disjoint subgraphs embed independently into that of H . Each connected union type of cycle rank at least $r + 1$ contributes $O(k^{-r})$, and there are finitely many, and so $E[X_H^2] = E[X_H](1 + o(1)) + E[X_H]^2(1 + o(1))$. The claim follows from $P(X_H \geq 1) \geq E[X_H]^2/E[X_H^2]$ and $P(X_H \geq 1) \leq E[X_H]$. $\square$

The second fact is that every occurrence produces its eigenvalue.

Proposition 5.5. *In the setting of Proposition 5.4, suppose that $r(H) \geq 2$ and that H binds. Then for every $\epsilon > 0$ there is a constant $c = c(B, H, \epsilon) > 0$ such that for all k sufficiently large,*

$$P(\lambda_{\text{new}}(H_k) \geq \lambda_B(H) - \epsilon) \geq c k^{1-r(H)}.$$

Proof. Let H_ℓ^* denote the truncation of H^* at depth ℓ from H . Since H^* is connected and locally finite, $\lambda_{\max}(H_\ell^*)$ increases to $\|A_{H^*}\|$ as $\ell \rightarrow \infty$, and so we may fix ℓ with $\lambda_{\max}(H_\ell^*) > \lambda_B(H) - \epsilon/2$. Let $u \geq 0$ be the Perron eigenvector of H_ℓ^* with $\|u\| = 1$, supported on $N = N(\ell)$ vertices. Call a copy of H in the cover *clean* if the depth ℓ neighborhood of its vertex set is isomorphic, over B , to H_ℓ^* . A copy which is not clean lies in a connected subgraph of the cover of size $O_\ell(1)$ and cycle rank at least $r(H) + 1$, since any deviation of the neighborhood from H_ℓ^* closes an additional independent cycle within distance ℓ , and the expected number of these is $O_\ell(k^{-r(H)})$ by the computation of Proposition 5.4, and so with probability at least $c_H k^{1-r(H)}(1 + o(1))$ the cover contains a clean copy.

Given a clean copy, transplant u onto its neighborhood. The subspace of functions which are constant on every fiber is invariant under the adjacency matrix A of the cover, its spectrum on this subspace is the spectrum of B , and its spectrum on the orthogonal complement is exactly the new spectrum. Writing P for the orthogonal projection onto the fiber constant functions, each fiber meets the support of u in at most N vertices, and so by the Cauchy-Schwarz inequality

$$\|Pu\|^2 = \sum_{b \in V(B)} \frac{1}{k} \left(\sum_{x \in \pi^{-1}(b)} u(x) \right)^2 \leq \frac{N}{k}.$$

Set $w = u - Pu$, a vector in the new subspace with $\|w\|^2 \geq 1 - N/k$. Since $u \geq 0$ and the neighborhood of the clean copy contains H_ℓ^* as a subgraph, we have that $\langle u, Au \rangle \geq \lambda_{\max}(H_\ell^*)$, and since $\|A\|$ is at most the maximum degree of B ,

$$\lambda_{\text{new}}(H_k) \geq \frac{\langle w, Aw \rangle}{\|w\|^2} \geq \langle u, Au \rangle - 3 \|A\| \|Pu\| \geq \lambda_{\max}(H_\ell^*) - O_\ell(k^{-1/2}),$$

whose right hand side is positive and exceeds $\lambda_B(H) - \epsilon$ for k large, and the result follows. $\square$

One family of defects is degenerate. A graph which is itself a cover of B , beginning with B under the identity, saturates every direction at every vertex, and so a copy of it is a full connected component, with defect eigenvalue its own Perron eigenvalue: its occurrence disconnects the cover, and by removal by value (subsection 2.4) the pipeline records instead the largest remaining eigenvalue outside the base spectrum. We exclude covers of B from the defect catalogue and from the minimum in the conjecture below, and by Proposition 5.4 the disconnected covers have probability $O(k^{1-r(B)})$. We believe

that the remaining defects account for the entire polynomial tail, and conjecture the matching upper bound.

Conjecture 5.6. *Let B be a connected base graph of cycle rank at least two, and let H_k be a uniformly random degree k cover of B , conditioned on connectedness. Then for every $\epsilon > 0$,*

$$P(\lambda_{\text{new}}(H_k) > \rho(B) + \epsilon) = k^{1-r(\epsilon)+o(1)},$$

where

$$r(\epsilon) = \min \{r(H) : H \text{ immerses in } B \text{ and is not a cover of } B, \lambda_B(H) > \rho(B) + \epsilon\},$$

and where the probability is smaller than every power of k when no such H exists.

For regular Ramanujan bases, Conjecture 5.6 is essentially the theorem of Friedman and Kohler [42, 41], and the coexistence of a Tracy-Widom bulk with a thin family of localized outliers is the picture proven at the sparse Erdős-Rényi edge [7, 8, 9, 55], and for sparse Wigner matrices the outliers are likewise produced by localized defects [101]. The eigenvectors near a Tracy-Widom edge are delocalized, and for random regular graphs they are locally Gaussian waves [51], as they are in small spectral windows for random lifts [32], so that the two populations differ in their eigenvectors as well as in their eigenvalues. The mechanism is not tied to randomness: Alon, Ganguly, and Srivastava [5] construct regular graphs of high girth whose top eigenvectors are fully localized on small vertex sets, and every value in $[2\sqrt{d-1}, d]$ is a limit point of second eigenvalues of d -regular graphs [6, 33]. Proposition 5.5 is the elementary half of Conjecture 5.6, and it already determines which moments diverge.

Corollary 5.7. *Suppose that some H immersing in B , not a cover of B , binds, and let $r^* \geq 2$ be the least cycle rank of such an H . Then*

$$E[|\lambda_{\text{new}}(H_k) - \rho(B)|^p] k^{2p/3} \rightarrow \infty$$

as $k \rightarrow \infty$ for every $p > 3(r^* - 1)/2$.

Proof. Let H bind with $r(H) = r^*$, and let $\Delta = \lambda_B(H) - \rho(B) > 0$. By Proposition 5.5 with $\epsilon = \Delta/2$, we have that $\lambda_{\text{new}}(H_k) - \rho(B) \geq \Delta/2$ with probability at least ck^{1-r^*} , and so

$$E[|\lambda_{\text{new}}(H_k) - \rho(B)|^p] k^{2p/3} \geq c \left(\frac{\Delta}{2}\right)^p k^{1-r^*+\frac{2p}{3}},$$

which tends to infinity exactly when $p > 3(r^* - 1)/2$. $\square$

Under Conjecture 1.6 the standard deviation is of order $k^{-2/3}$, and so a defect at height Δ sits at $\Delta k^{2/3}$ standard deviations above the mean, and it contributes to the p^{th} standardized moment at order

$$(5.1) \quad k^{1-r^*} (\Delta k^{2/3})^p \asymp k^{1-r^*+\frac{2p}{3}}.$$

Under Conjecture 5.6, holding uniformly down to the fluctuation scale, the moments below the threshold of Corollary 5.7 converge instead to those of the limit law. Equation (5.1) organizes every moment anomaly in this paper. For the permutation multigraph model and for $K_4 - e$, where $r^* = 2$, the variance itself is eventually dominated by the defect band, growing relative to the bulk like $k^{1/3}$, and at the bulk scale the skewness diverges like k and the kurtosis like $k^{5/3}$, so that the raw moments of these families are the most contaminated in the dataset, which is what subsection 3.4 observed, and the raw ratio g_V of subsection 5.3 must eventually turn around and decrease, with only its trimmed version converging to the TW_1 ratio. For $K_5 - e$ and K_5 , where $r^* = 3$,

the variance is safe, the kurtosis diverges like $k^{2/3}$, matching the observed growth, and $p = 3$ is exactly critical: the defects add a constant, non-vanishing offset to the skewness, which is why the $K_5 - e$ skewness settles near 0.36 rather than at the Tracy-Widom value 0.293, while for K_5 the constant is too small to resolve. Every quantile level statistic is immune, since the total defect mass vanishes: the Kolmogorov-Smirnov distance moves by at most the tail mass and the Ramanujan probability by less, and this is why the fixed base families are omitted from Figure 4.4 while every distributional test of subsection 5.1 converges.

The data confirms the computed defects in detail. We evaluated $\lambda_B(H)$ by the branch recursion, and confirmed the values by planting the subgraphs in sampled covers. For $K_5 - e$, no subgraph of cycle rank two binds, the complete graph K_4 , of cycle rank three, binds at

$$\lambda_{K_5-e}(K_4) = 3.3391340785\dots,$$

with its subdivisions filling the interval below it, and the largest cycle rank four value is $3.4519402\dots$, while a disjoint copy of $K_5 - e$ itself would give $1 + \sqrt{7}$. For K_5 the same computation gives $\lambda_{K_5}(K_4) = 7/2$ exactly.

Numerical Result 5.8. The far outliers of the $K_5 - e$ family at the largest sizes fill an interval above ρ , leave a gap, and cluster on the K_4 atom $3.33913\dots$, where the extreme samples at $V = 5000$ and $10,000$ sit, and the extremes at $V \leq 2000$ sit instead on the rank four atom, which decays faster and is gone by $V = 5000$. The count in a window around the K_4 atom decays with fitted exponent 2.02 ± 0.13 against the predicted k^{-2} , and at thresholds beyond the reach of cycle rank three the fitted exponent is 2.90 ± 0.14 against k^{-3} . For the K_5 family, the six samples more than nine standard deviations above the mean at $V = 10,000$ all lie in $[3.497, 3.510]$, around the atom $7/2$.

The apparent lengthening of the $K_5 - e$ tail, from 27 standard deviations at $V = 5000$ to 42 at $V = 10,000$, is then exact: a fixed atom measured in units of $\sigma_V \propto V^{-2/3}$ lengthens by $2^{2/3} = 1.587$, and the observed ratio is 1.584 (Figure 5.6). Table 5.1 collects the catalogue across all four families with binding defects of low rank.

5.3. The Boundary Case $K_4 - e$. The third fixed base, $K_4 - e$, tests Conjecture 1.6 at its boundary: the complete graph on four vertices minus an edge has cycle rank exactly two, degrees $(2, 3, 3, 2)$, and the closed form

$$\rho(K_4 - e) = \sqrt{1 + 2\sqrt{7}} = 2.5082867902\dots$$

Suppressing its two degree two vertices identifies $K_4 - e$ with the theta graph, two vertices joined by three parallel edges, and so random covers of $K_4 - e$ are subdivisions of random theta covers, which are exactly the bipartite cubic multigraphs built from three independent uniform permutations, the bipartite model of MNS and of Novikoff's experiments [90]. We sampled 10^6 covers per size at the same seven sizes $V = 4k$ between 100 and 10,000.

Numerical Result 5.9. The covers of $K_4 - e$ converge more slowly than any other family. At $V = 10,000$ the ratio g_V is 0.7326 and the Ramanujan probability 0.79490, both rising toward their conjectured limits. The fitted amplitude descends through 0.2382, 0.2275, and 0.2219 at $V = 2000, 5000$, and $10,000$, and extrapolates to 0.214 to 0.216, against the predicted $C_B^{-2/3} = 0.212519$. After the affine fit the Kolmogorov-Smirnov distance to TW_1 is 2.8×10^{-3} , against 3.8×10^{-3} for TW_2 and 5.9×10^{-3} for TW_4 , but also against 2.6×10^{-3} for the maximum law $\max(TW_2, TW'_2)$, which is not

Far outliers lock onto the localized defect eigenvalues (dotted); frequencies decay like k^{1-r}

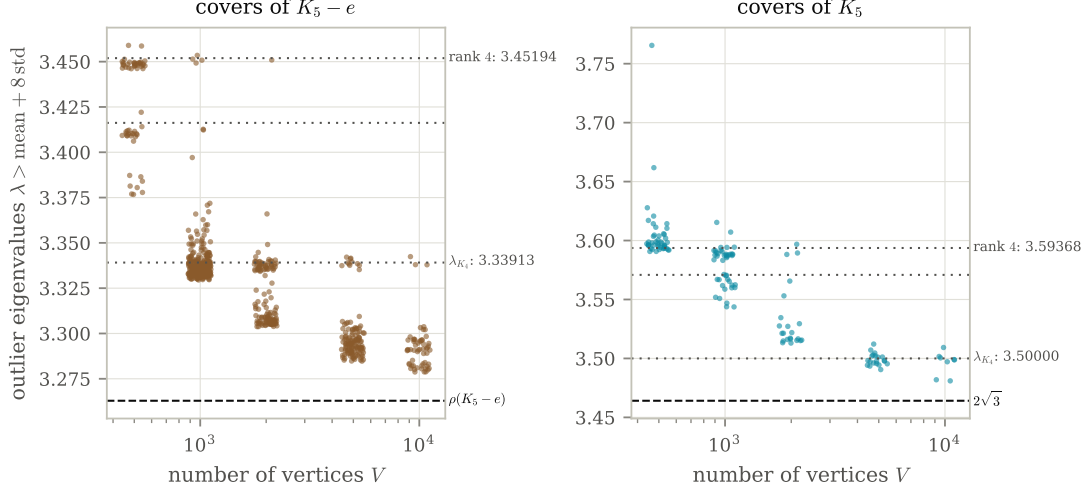


FIGURE 5.6. Every sample more than eight standard deviations above the mean, against V , for the $K_5 - e$ (left) and K_5 (right) cover families, with the computed defect eigenvalues overlaid: the outliers lock onto the K_4 atoms, $3.33913\dots$ for $K_5 - e$ and exactly $7/2$ for K_5 , and onto the cycle rank four values (dotted), while the continuum of K_4 subdivisions drains toward ρ and the frequencies decay like k^{1-r} .

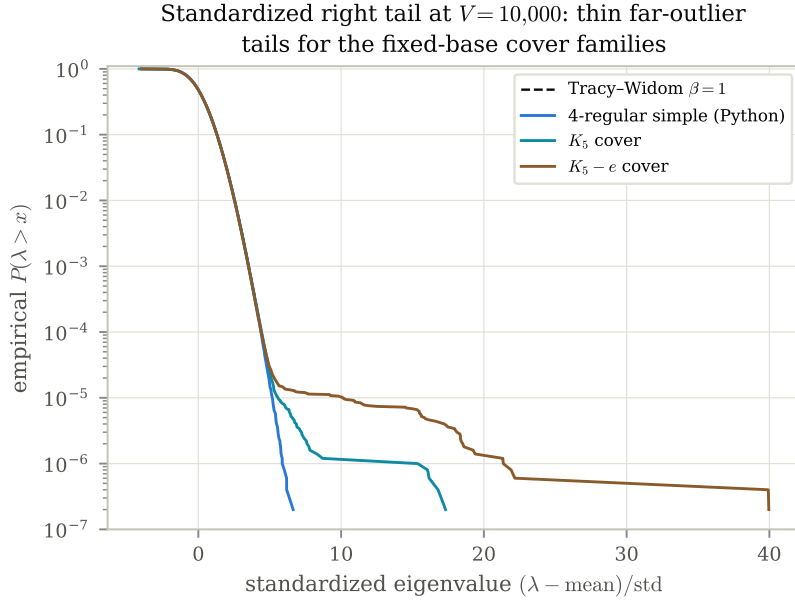


FIGURE 5.7. Standardized right tail $P(\lambda > x)$ at $V = 10,000$ for the simple 4-regular, K_5 cover, and $K_5 - e$ cover families. The bulk of all three collapses onto the Tracy-Widom $\beta = 1$ tail, while the $K_5 - e$ family carries a thin far outlier tail at frequency near 10^{-5} , reaching 42 standard deviations, and the K_5 family a much thinner one.

base B	defect H	$r(H)$	$\lambda_B(H)$	rate k^{1-r}	fitted exponent
bouquet B_2	triple edge, adjacent loops	2	7/2	V^{-1}	too few to fit
bouquet B_2	loop on a double edge	2	3.5385847...	V^{-1}	too few to fit
bouquet B_2	triangle with a loop	2	3.4675039...	V^{-1}	too few to fit
$K_5 - e$	K_4	3	3.3391340785...	k^{-2}	2.02 ± 0.13
$K_5 - e$	largest rank four	4	3.4519402...	k^{-3}	2.90 ± 0.14
K_5	K_4	3	7/2	k^{-2}	1.7 ± 0.7
$K_4 - e$	$\theta(2, 2, 4)$	2	2.5118521...	k^{-1}	≈ 1.3 , descending
$K_4 - e$	$\theta(1, 2, 5)$	2	2.5136918...	k^{-1}	≈ 1.3 , descending
$K_4 - e$	rank three band	3	$\lesssim 2.530$	k^{-2}	1.96

TABLE 5.1. The catalogue of leading localized defects. Each row lists a subgraph H immersing in B , its cycle rank $r(H)$, the defect eigenvalue $\lambda_B(H)$ of Definition 5.3 computed by the branch recursion, the occurrence rate $k^{1-r(H)}$ of Proposition 5.4, and the exponent fitted to the observed outlier counts. The bouquet rows are the permutation multigraph model of subsection 3.4, with $k = V$, where 18 samples across four sizes are too few for a fitted exponent. Here $\theta(a, b, c)$ is the generalized theta graph whose two branch vertices are joined by routes of lengths a , b , and c , and the two $K_4 - e$ atoms sit too close together for separate counts, so one exponent is fitted to the pair.

a candidate for this family and whose shape is close to that of TW_1 . The standardized sample, after discarding the 0.2% of samples beyond six standard deviations, is nearest $\widetilde{TW}_1$ from $V = 1000$ on, at distance 4.0×10^{-3} at $V = 10,000$ against 6.4×10^{-3} and 9.9×10^{-3} for $\widetilde{TW}_2$ and $\widetilde{TW}_4$.

Two separate causes slow the family down. The first is the scale. The square root edge of μ_{K_4-e} is steep, $C_B = 10.207$, and narrow: $\pi \mu'_B(x)/\sqrt{\rho - x}$ is within 3% of C_B only within about 10^{-3} of the edge, against about 10^{-2} for $K_5 - e$ (Figure 2.3), and the fluctuation scale $C_B^{-2/3}V^{-2/3}$ enters this window only at $V \approx 3000$. The second cause is the far tail. The raw distributional statistics are contaminated in a way the other covers only hint at: the excess kurtosis grows to 12.7, and the Kolmogorov-Smirnov distance of the standardized sample to $\widetilde{TW}_1$, after falling to 3.8×10^{-3} at $V = 500$, grows again to 1.9×10^{-2} at $V = 10,000$ (Figure 5.8, left).

The tail itself differs from the $K_5 - e$ tail exactly as the defect picture predicts. No cycle rank one subgraph binds, and no connected 2-cover of $K_4 - e$ has a new eigenvalue above ρ either, as a direct check of all 32 signings shows, however the base now has cycle rank two, and so proper rank two subgraphs exist, generalized thetas whose routes are longer than the base's and which carry branches, occurring at the rate k^{-1} of Proposition 5.4. The two shortest, $\theta(2, 2, 4)$ and $\theta(1, 2, 5)$ in the notation of Table 5.1, bind at

$$2.5118521\dots \quad \text{and} \quad 2.5136918\dots,$$

computed by the same branch recursion, one power of k more frequent than the leading $K_5 - e$ defects, and above the rank two band, rank three shapes reach ≈ 2.530 at rate k^{-2} . The data shows precisely this structure (Figure 5.8, right): the outliers form a dense band draining toward ρ rather than isolated atoms, capped near 2.530 at the largest sizes, with the count above the rank two cap decaying with fitted exponent 1.96 and the count at the cap decaying visibly more slowly, still descending toward the predicted 1. In the notation of Conjecture 5.6, $r(\epsilon) = 2$ for ϵ below ≈ 0.006 and 3 up

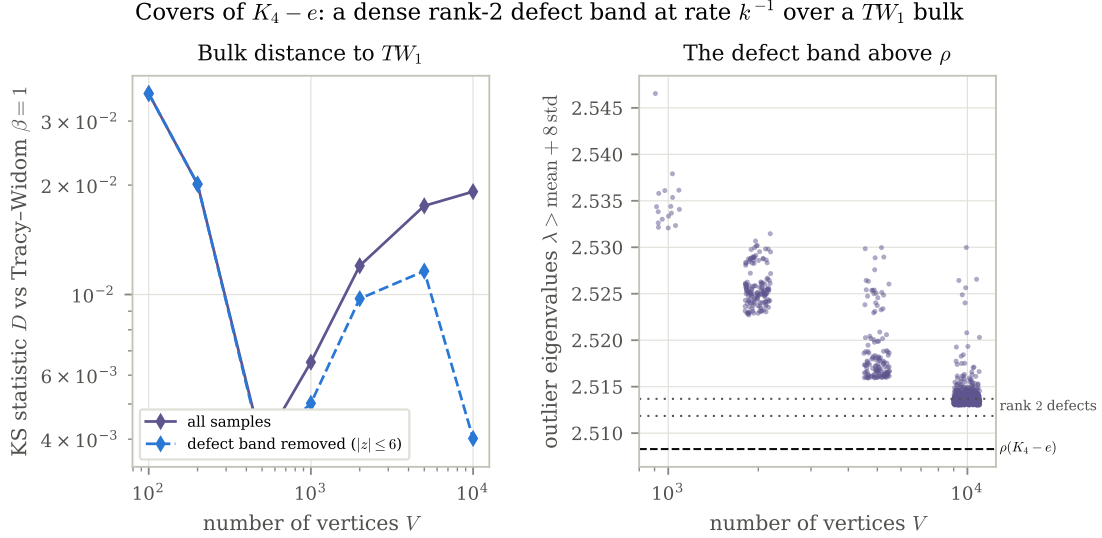


FIGURE 5.8. Covers of $K_4 - e$. Left: Kolmogorov-Smirnov distance of the standardized sample to $\widetilde{TW}_1$, for all samples and with the defect band removed. Right: every sample more than eight standard deviations above the mean against V , with $\rho(K_4 - e)$ dashed and the two computed rank two defect eigenvalues dotted. The band drains toward ρ at rate k^{-1} , with rank three shapes reaching ≈ 2.530 at rate k^{-2} .

to ≈ 0.022 . The comparison across the three fixed bases is the point: the denser the base, the higher the cycle rank at which the defect hierarchy starts, and the further the atoms sit from ρ . For $K_4 - e$ the hierarchy starts at rank two, the band crowds the bulk at accessible sizes, and the moments and even the raw Kolmogorov-Smirnov distance feel it, which together with its narrow square root edge is why this family, at the boundary of the conjecture, converges last.

6. OPEN PROBLEMS

In this section we discuss the problems that the data leaves open.

The Block Laws. Conjecture 1.3 is open in every case, including $\beta = 1$: for covers of a d -regular base of size $n \geq 2$ no analog of Theorem 1.1 is known, and for irregular bases even the correct formulation of edge rigidity around $\rho(B)$ is open. The abelian case seems the natural first target, since the block A_χ is an explicit random Hermitian matrix built from a random regular graph and independent uniform phases, and the loop equation approach, which characterizes the Airy_β statistics for every rational β and has already given a simplified proof of Theorem 1.1 [23], seems well suited to it (see also [62]).

Problem 6.1. Prove that the largest eigenvalue of the $\mathbb{Z}_3$ -character block A_χ of a random $\mathbb{Z}_3$ -cover of a random simple d -regular graph has TW_2 fluctuations, in the sense of Conjecture 1.3.

The Scale and the Location. All of the scale constants in our data are predicted by the square root edge of the limiting spectral measure, with the Gaussian ensembles normalized alike, and the prediction is confirmed to within 0.15% after extrapolation for K_5 and $K_5 - e$. The location is a different matter. The shift a/n of (2.6) has a

family dependent constant, from 4.47 for the simple 4-regular graphs to -1.44 for the multigraph and nearly 0 for the covers of K_5 , and it controls the approach of every Ramanujan probability to its limit, at rate $n^{-1/3}$.

Problem 6.2. Determine the location coefficient a . In particular, is a determined by the expected numbers of short cycles of the model, as the $1/n$ corrections to the Kesten-McKay density computed by Metz, Parisi, and Leuzzi [83] suggest, and does the difference of about 6 between the simple and multigraph models come from the conditioning on the absence of loops and double edges?

I believe that the shift is a deterministic consequence of the short cycles, since in the families of our data it changes when the cycle statistics are changed and not otherwise, but I do not know how to compute it.

Far Outliers. The permutation multigraph model of subsection 3.4 has the Tracy-Widom bulk of the simple model together with a thin tail of defects of cycle rank two, and the fixed base families of section 5 carry tails of the same character.

Problem 6.3. Prove that the largest non-trivial eigenvalue of the permutation multigraph model, rescaled as in Theorem 1.1, converges to TW_1 in distribution. Is the point process of its eigenvalues above $2\sqrt{3} + \epsilon$ asymptotically a Poisson process on the three defect eigenvalues of subsection 3.4, with total intensity of order $1/V$?

Problem 6.4. Prove the upper bound in Conjecture 5.6: that nothing but the defects of subsection 5.2 inhabits the polynomial tail. The lower bound is Proposition 5.5, the regular Ramanujan case is the theorem of Friedman and Kohler [42, 41], and the fitted exponents of Numerical Results 5.8 and 5.9 match the conjectured rates on two irregular bases with different leading ranks.

Growing Covers. Huang and Yau [61] proved Tracy-Widom fluctuations for regular graphs of growing degree. The analogous regime for covers, base size fixed and cover group growing, is precisely the setting of Conjecture 1.6, however one may also grow the base: for $\mathbb{Z}_k$ -covers with k growing with n , the number of blocks of class $\beta = 2$ grows, and the largest new eigenvalue becomes the maximum of a growing number of Tracy-Widom variables. The maximum of independent Tracy-Widom variables lies in the Gumbel domain of attraction [29], which suggests a transition from Tracy-Widom to Gumbel fluctuations in the spirit of [64], and the Ramanujan probability $F_2(0)^{(k-1)/2}$ of Conjecture 1.4 then tends to 0, in line with the theorem of Agarwal, Chandrasekaran, Kolla, and Madan [2] that abelian lifts of exponentially large degree are never close to Ramanujan. For iterated 2-covers of dense random regular graphs, Liu and Zou [73] prove exactly this maximum of independent Tracy-Widom structure.

Problem 6.5. For random $\mathbb{Z}_k$ -covers of a random d -regular graph on n vertices, determine the joint regimes of (k, n) in which the largest new eigenvalue has Tracy-Widom, and Gumbel, fluctuations.

Ramanujan Covers. Covers also carry the constructive side of the Ramanujan story. Lubotzky, Phillips, and Sarnak [75] and independently Margulis [80] built infinite Ramanujan families when $d - 1$ is prime, and whether non-bipartite Ramanujan graphs exist for every degree remains open, though explicit near Ramanujan graphs of every degree and size are known [4]. Bilu and Linial [15] conjectured that every d -regular graph has a 2-cover whose new eigenvalues all lie in $[-2\sqrt{d-1}, 2\sqrt{d-1}]$, which for Ramanujan base graphs would settle it. A counterexample among general cubic graphs

has recently been announced by Xu [107], while the case of Ramanujan base graphs, which would suffice, remains open. Marcus, Spielman, and Srivastava [79] proved the one-sided statement for bipartite graphs, Hall, Puder, and Sawin [49] proved that every connected loopless graph has a one-sided Ramanujan cover of every degree, and Chandrasekaran and Velingker [26] constructed shift lifts of degrees three and four preserving the Ramanujan property. Conjectures 1.4 and 1.6 assert far more than existence, namely that a positive proportion of random covers is one-sided Ramanujan. For the 2-covers of a random regular graph, whose single new block is the real block of the sign character, that proportion is $F_1(0)$, and since the randomness of the base is invisible at the fluctuation scale in our data (subsection 2.4), the proportion should be the same for almost every base graph.

Problem 6.6. Let G be a uniformly random simple d -regular graph on n vertices. Prove that, with probability tending to 1 as $n \rightarrow \infty$, a proportion bounded below by a positive constant of the $2^{e(G)}$ signings of G have all of their eigenvalues at most $2\sqrt{d-1}$. Does this proportion converge in probability to $F_1(0) = 0.8319081\dots$?

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